



NET Benchmark

MCQ STRATEGY BOOK

Not just prep. The prep.

A concept-first guide to MCQ patterns, traps, shortcuts, and answer choices.

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Mathematics Physics English

Final test is reserved separately at the end.



I believe Allah is the ultimate tester

Not just prep. The *prep.*

NET Benchmark

MCQ Strategy Book

by Dr. BASHIR

This version is not a practice-test dump. It is arranged as a preparation book. Each MCQ is treated as a learning case: the route, the trap, the answer, the reasoning, the wrong-option analysis, and a practical tip.

BENCHMARK

Not just prep. The prep.

Use the final test only after completing the preparation chapters.



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How to Use This Book

This book is designed for preparation rather than testing. Each MCQ is treated as a compact lesson so that the learner studies the pattern, not only the answer.

Study method. Attempt the MCQ first. Then read the Fast Route. After that, read the explanation and the wrong-option analysis. Finish by writing one personal note from the Trick / Tip section.

Goal. The aim is to recognize repeatable MCQ structures: the formula being tested, the tempting wrong answer, and the shortest reliable method.

Final test. The actual final test is kept as a separate ending section and should not be mixed with the teaching pages.

Not just prep. The prep.

Book Positioning

The book is built from the 200-question bank, but the treatment is instructional: each MCQ includes wrong-answer explanations and targeted strategy notes.



CHAPTER 1

Mathematics MCQ Strategy Lab

How to use this chapter. Use each page to reduce the problem to the tested mathematical rule. The wrong answers usually come from a sign error, missed condition, or wrong formula.



Q001

Mathematics Complex Numbers

MCQ Focus. If $z = (3 - 4i)/(1 + 2i)$, the real part of z is:

- A. -1
- B. 1
- C. -5
- D. $\frac{11}{5}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A -1

Worked Reasoning. Rationalize the denominator: multiply by the conjugate $1-2i$. The expression becomes $(-5-10i)/5$, so the real part is -1.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from -1.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from -1.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from -1.

Trick / Tip. The useful shortcut here is tied to complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with -1, not just look numerically close.

Q002

Mathematics Complex Numbers

MCQ Focus. The modulus of $(1 + i)^6$ is:

- A. 4
- B. 8
- C. 16
- D. 64

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 8

Worked Reasoning. Use $|zw| = |z||w|$. Since $|1+i| = \sqrt{2}$, $|(1+i)^6| = (\sqrt{2})^6 = 8$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 8.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 8.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 8.

Trick / Tip. For this Complex Numbers pattern, notice complex numbers: When ω appears, immediately write $1+\omega+\omega^2=0$ and $\omega^3=1$ before touching the options.. The correct option should agree with 8, not just look numerically close.



Q003

Mathematics Complex Numbers

MCQ Focus. If ω is a non-real cube root of unity, then $1 + \omega^7 + \omega^8$ equals:

- A. 0
- B. 1
- C. ω
- D. ω^2

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A 0

Worked Reasoning. For a cube root of unity, $\omega^3=1$ and $1+\omega+\omega^2=0$. Reduce powers: $\omega^7=\omega$ and $\omega^8=\omega^2$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 0.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 0.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 0.

Trick / Tip. This question rewards recognising cube-root identity: write ω cubed as 1 and $1+\omega+\omega$ squared as 0 before simplifying. The correct option should agree with 0, not just look numerically close.

Q004

Mathematics Complex Numbers

MCQ Focus. The value of i^{137} is:

- A. i
- B. $-i$
- C. 1
- D. -1

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A i

Worked Reasoning. Powers of i repeat every 4. Since 137 leaves remainder 1 on division by 4, $i^{137} = i$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from i .
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from i .
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from i .

Trick / Tip. Question 4 is a power cycle item: reduce the exponent modulo 4 for powers of i , then read the sign from the cycle 1, i , -1 , $-i$. The correct option should agree with i , not just look numerically close.



Q005

Mathematics

Mathematics Pattern Recognition

MCQ Focus. If $|x - 2| < 3$, then x belongs to:

- A. $(-1,5)$
- B. $(-5,1)$
- C. $(2,3)$
- D. $(-3,2)$

Fast Route. Spot the tested rule first and avoid long working.

Trap Watch. The distractor usually uses the right topic but the wrong rule.

Answer. A $(-1,5)$

Worked Reasoning. $|x - 2| < 3$ means $-3 < x - 2 < 3$. Adding 2 gives $-1 < x < 5$.

Why the other options are wrong.

- ◇ Option B: uses a nearby but wrong pattern; it leads away from $(-1,5)$.
- ◇ Option C: conflicts with the calculated result; it leads away from $(-1,5)$.
- ◇ Option D: does not follow the rule used in the worked solution; it leads away from $(-1,5)$.

Trick / Tip. The useful shortcut here is tied to absolute-value interval: split the inequality into a compound interval and solve both sides at once. The correct option should agree with $(-1,5)$, not just look numerically close.

Q006

Mathematics Complex Numbers

MCQ Focus. For sets A and B, $n(A) = 30$, $n(B) = 25$ and $n(A \cup B) = 45$. Then $n(A \text{ intersection } B)$ is:

- A. 5
- B. 10
- C. 15
- D. 20

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 10

Worked Reasoning. Use $n(A \text{ union } B) = n(A) + n(B) - n(A \text{ intersection } B)$. Therefore $45 = 30 + 25 - x$, so $x = 10$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 10.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 10.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 10.

Trick / Tip. For this Complex Numbers pattern, notice set inclusion-exclusion: use $n(A \text{ union } B) = n(A) + n(B) - n(A \text{ intersection } B)$ and solve for the missing region. The correct option should agree with 10, not just look numerically close.



Q007

Mathematics Complex Numbers

MCQ Focus. If $f(x) = 2x - 3$ and $g(x) = x^2 + 1$, then $(g \circ f)(2)$ is:

- A. 0
- B. 2
- C. 5
- D. 10

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 2

Worked Reasoning. First find $f(2) = 1$, then apply g : $g(1) = 1^2 + 1 = 2$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 2.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 2.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 2.

Trick / Tip. This question rewards recognising function rule: substitute carefully and respect the order of composition or inversion. The correct option should agree with 2, not just look numerically close.

Q008

Mathematics Sets and Functions

MCQ Focus. If $f(x) = x/(x-1)$, then $f(f(x))$ equals:

- A. x
- B. $1/x$
- C. $x-1$
- D. $1-x$

Fast Route. Write the defining rule or restriction in one line; the answer usually follows without full expansion.

Trap Watch. Do not skip exclusions; domains and set differences often hide the missing restriction.

Answer. A x

Worked Reasoning. Substitute $f(x)$ into f again. The algebra simplifies back to x .

Why the other options are wrong.

- ◇ Option B: reverses the order of composition, inverse, or set difference; it leads away from x .
- ◇ Option C: counts the wrong region or keeps an element that should be removed; it leads away from x .
- ◇ Option D: misses an exclusion or restriction in the domain/set condition; it leads away from x .

Trick / Tip. Question 8 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with x , not just look numerically close.



Q009

Mathematics Complex Numbers

MCQ Focus. The inverse of $f(x) = 3x + 5$ is:

- A. $(x-5)/3$
- B. $3x-5$
- C. $(x+5)/3$
- D. $5x+3$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $(x-5)/3$

Worked Reasoning. Set $y=3x+5$ and solve for x : $x=(y-5)/3$. Replace y by x .

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $(x-5)/3$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $(x-5)/3$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $(x-5)/3$.

Trick / Tip. The useful shortcut here is tied to complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $(x-5)/3$, not just look numerically close.



Q010

Mathematics

Complex Numbers

MCQ Focus. The domain of $f(x) = \sqrt{x-2}/(x-5)$ is:

- A. $[2, \text{infinity})$
- B. $[2,5) \cup (5, \text{infinity})$
- C. $\{5\}$
- D. $\{1, 2, 5\}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** $[2,5) \cup (5, \text{infinity})$

Worked Reasoning. The square root requires $x \geq 2$, and the denominator requires x not equal to 5.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $[2,5) \cup (5, \text{infinity})$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $[2,5) \cup (5, \text{infinity})$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $[2,5) \cup (5, \text{infinity})$.

Trick / Tip. For this Complex Numbers pattern, notice complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $[2,5) \cup (5, \text{infinity})$, not just look numerically close.



Q011

Mathematics Complex Numbers

MCQ Focus. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$, then $A-B$ is:

- A. $\{1, 2\}$
- B. $\{3, 4\}$
- C. $\{5\}$
- D. $\{1, 2, 5\}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $\{1, 2\}$

Worked Reasoning. $A-B$ means elements in A that are not in B . Remove 3 and 4 from A .

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\{1, 2\}$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\{1, 2\}$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\{1, 2\}$.

Trick / Tip. This question rewards recognising complex numbers: When ω appears, immediately write $1+\omega+\omega^2=0$ and $\omega^3=1$ before touching the options.. The correct option should agree with $\{1, 2\}$, not just look numerically close.

Q012

Mathematics Complex Numbers

MCQ Focus. The determinant of $\begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$ is:

- A. -1
- B. 1
- C. 29
- D. -29

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A -1

Worked Reasoning. For a 2 by 2 matrix, determinant is $ad-bc$. Here $2(7)-3(5)=-1$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from -1.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from -1.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from -1.

Trick / Tip. Question 12 is a determinant item: for a 2 by 2 matrix, compute $ad-bc$; the cross-product order decides the sign. The correct option should agree with -1, not just look numerically close.



Q013

Mathematics Complex Numbers

MCQ Focus. If A is a 2 by 2 matrix with $\det(A)=3$, then $\det(2A)$ is:

- A. 6
- B. 9
- C. 12
- D. 24

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 12

Worked Reasoning. For a 2 by 2 matrix, $\det(kA)=k^2 \det(A)$. Thus $\det(2A)=4(3)=12$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 12.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 12.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 12.

Trick / Tip. The useful shortcut here is tied to determinant: for a 2 by 2 matrix, compute $ad-bc$; the cross-product order decides the sign. The correct option should agree with 12, not just look numerically close.



Q014

Mathematics Complex Numbers

MCQ Focus. For matrix $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, the sum of diagonal elements is:

- A. 4
- B. 5
- C. 6
- D. 10

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 5

Worked Reasoning. The diagonal entries are 1 and 4, and their sum is 5.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 5.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 5.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 5.

Trick / Tip. For this Complex Numbers pattern, notice matrix property: name the matrix property first - determinant, trace, rank, inverse, or order - because each uses a different rule. The correct option should agree with 5, not just look numerically close.



Q015

Mathematics Complex Numbers

MCQ Focus. If A is singular, then:

- A. $\det(A)=0$
- B. $\det(A)=1$
- C. A has no zero entries
- D. A must be diagonal

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $\det(A)=0$

Worked Reasoning. A singular matrix has no inverse; equivalently, its determinant is zero.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\det(A)=0$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\det(A)=0$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\det(A)=0$.

Trick / Tip. This question rewards recognising determinant: for a 2 by 2 matrix, compute $ad-bc$; the cross-product order decides the sign. The correct option should agree with $\det(A)=0$, not just look numerically close.

Q016

Mathematics Complex Numbers

MCQ Focus. The inverse of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ has determinant:

- A. $-\frac{1}{2}$
- B. $\frac{1}{2}$
- C. -2
- D. 2

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $-\frac{1}{2}$

Worked Reasoning. $\det(A^{-1}) = 1/\det(A)$. Here $\det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = -2$, so the inverse determinant is $-\frac{1}{2}$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $-\frac{1}{2}$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $-\frac{1}{2}$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $-\frac{1}{2}$.

Trick / Tip. Question 16 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $-\frac{1}{2}$, not just look numerically close.



Q017

Mathematics Complex Numbers

MCQ Focus. If A is 2 by 3 and B is 3 by 4, then AB has order:

- A. 2 by 4
- B. 3 by 3
- C. 4 by 2
- D. Not defined

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A 2 by 4

Worked Reasoning. Matrix multiplication (2 by 3)(3 by 4) is defined and gives a 2 by 4 matrix.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 2 by 4.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 2 by 4.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 2 by 4.

Trick / Tip. The useful shortcut here is tied to complex numbers: When ω appears, immediately write $1+\omega+\omega^2=0$ and $\omega^3=1$ before touching the options.. The correct option should agree with 2 by 4, not just look numerically close.

Q018

Mathematics Complex Numbers

MCQ Focus. If $\det \begin{pmatrix} x & 2 \\ 8 & x \end{pmatrix} = 0$, then x is:

- A. 4 only
- B. -4 only
- C. ± 4
- D. 0

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C ± 4

Worked Reasoning. The determinant is $x^2 - 16$. Setting it to zero gives $x = \pm 4$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from ± 4 .
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from ± 4 .
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from ± 4 .

Trick / Tip. For this Complex Numbers pattern, notice complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with ± 4 , not just look numerically close.



Q019

Mathematics Complex Numbers

MCQ Focus. The rank of $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ is:

- A. 0
- B. 1
- C. 2
- D. 4

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 1

Worked Reasoning. The second row is twice the first row, so there is only one independent row.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 1.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 1.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 1.

Trick / Tip. This question rewards recognising rank/dependence: look for proportional rows or columns; dependence usually lowers rank immediately. The correct option should agree with 1, not just look numerically close.

Q020

Mathematics Complex Numbers

MCQ Focus. For $x^2 - 5x + 6 = 0$, the product of roots is:

- A. -5
- B. 5
- C. 6
- D. -6

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 6

Worked Reasoning. For $ax^2 + bx + c = 0$, product of roots is c/a . Here it is 6.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 6.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 6.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 6.

Trick / Tip. Question 20 is a quadratic structure item: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with 6, not just look numerically close.



Q021

Mathematics Quadratics and Algebra

MCQ Focus. If the roots of $x^2 - kx + 16 = 0$ are equal, k equals:

- A. 4
- B. 8
- C. ± 8
- D. 16

Fast Route. Use Vieta, discriminant, substitution, or cover-up before attempting a long solution.

Trap Watch. Sign errors and confusing sum/product of roots create most wrong options.

Answer. C ± 8

Worked Reasoning. Equal roots require discriminant zero: $k^2 - 64 = 0$, so $k = \pm 8$.

Why the other options are wrong.

- ◇ Option A: uses the sum of roots where the product or substitution condition is being tested; it leads away from ± 8 .
- ◇ Option B: comes from a sign slip in the coefficient or discriminant step; it leads away from ± 8 .
- ◇ Option D: uses the sum of roots where the product or substitution condition is being tested; it leads away from ± 8 .

Trick / Tip. The useful shortcut here is tied to complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with ± 8 , not just look numerically close.

Q022

Mathematics Quadratics and Algebra

MCQ Focus. The roots of $x^2 + 4x + 13 = 0$ are:

- A. $-2 \pm 3i$
- B. $2 \pm 3i$
- C. $-4 \pm 13i$
- D. No roots

Fast Route. Use Vieta, discriminant, substitution, or cover-up before attempting a long solution.

Trap Watch. Sign errors and confusing sum/product of roots create most wrong options.

Answer. **A** $-2 \pm 3i$

Worked Reasoning. Apply the quadratic formula. The discriminant is $16 - 52 = -36$, giving $-2 \pm 3i$.

Why the other options are wrong.

- ◇ Option B: comes from a sign slip in the coefficient or discriminant step; it leads away from $-2 \pm 3i$.
- ◇ Option C: stops before the algebraic simplification that the options require; it leads away from $-2 \pm 3i$.
- ◇ Option D: uses the sum of roots where the product or substitution condition is being tested; it leads away from $-2 \pm 3i$.

Trick / Tip. For this Quadratics and Algebra pattern, notice complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $-2 \pm 3i$, not just look numerically close.



Q023

Mathematics Complex Numbers

MCQ Focus. If α and β are roots of $2x^2 - 7x + 3 = 0$, $\alpha + \beta$ is:

- A. $\frac{7}{2}$
- B. $-\frac{7}{2}$
- C. $\frac{3}{2}$
- D. $-\frac{3}{2}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $\frac{7}{2}$

Worked Reasoning. Sum of roots is $-b/a$. For $2x^2 - 7x + 3$, this is $\frac{7}{2}$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\frac{7}{2}$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\frac{7}{2}$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{7}{2}$.

Trick / Tip. This question rewards recognising complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\frac{7}{2}$, not just look numerically close.

Q024

Mathematics Complex Numbers

MCQ Focus. If one root of $x^2 - 6x + k = 0$ is 2, then k is:

- A. 4
- B. 6
- C. 8
- D. 12

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 8

Worked Reasoning. Substitute $x=2$ into $x^2 - 6x + k = 0$: $4 - 12 + k = 0$, hence $k=8$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 8.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 8.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 8.

Trick / Tip. Question 24 is a quadratic structure item: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with 8, not just look numerically close.

Q025

Mathematics Complex Numbers

MCQ Focus. For $x^2 + px + q = 0$, if roots are 3 and -5, then $p+q$ equals:

- A. -17
- B. -13
- C. 13
- D. 17

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B -13

Worked Reasoning. If roots are 3 and -5, their sum is -2 and product is -15. Thus $p=-2$, $q=-15$, and $p + q = -17$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from -13.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from -13.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from -13.

Trick / Tip. The useful shortcut here is tied to quadratic structure: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with -13, not just look numerically close.

Q026

Mathematics Complex Numbers

MCQ Focus. The quadratic whose roots are reciprocals of roots of $3x^2 - 5x + 2 = 0$ is:

- A. $2x^2 - 5x + 3 = 0$
- B. $3x^2 + 5x + 2 = 0$
- C. $x^2 - 5x + 6 = 0$
- D. $5x^2 - 3x + 2 = 0$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $2x^2 - 5x + 3 = 0$

Worked Reasoning. For reciprocal roots, reverse the coefficients of the original quadratic: $2x^2 - 5x + 3 = 0$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $2x^2 - 5x + 3 = 0$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $2x^2 - 5x + 3 = 0$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $2x^2 - 5x + 3 = 0$.

Trick / Tip. For this Complex Numbers pattern, notice quadratic structure: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with $2x^2 - 5x + 3 = 0$, not just look numerically close.



Q027

Mathematics Mathematics Pattern Recognition

MCQ Focus. If $1/[x(x+2)] = A/x + B/(x+2)$, then $A-B$ equals:

- A. 0
- B. 1
- C. -1
- D. 2

Fast Route. Spot the tested rule first and avoid long working.

Trap Watch. The distractor usually uses the right topic but the wrong rule.

Answer. **B** 1

Worked Reasoning. Set $x=0$ to get $A=\frac{1}{2}$ and set $x=-2$ to get $B=-\frac{1}{2}$; hence $A-B = 1$.

Why the other options are wrong.

- ◇ Option A: does not follow the rule used in the worked solution; it leads away from 1.
- ◇ Option C: conflicts with the calculated result; it leads away from 1.
- ◇ Option D: does not follow the rule used in the worked solution; it leads away from 1.

Trick / Tip. This question rewards recognising complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 1, not just look numerically close.

Q028

Mathematics Complex Numbers

MCQ Focus. If $(3x+5)/[(x+1)(x+2)] = A/(x+1)+B/(x+2)$, then A is:

- A. -2
- B. 2
- C. 5
- D. 8

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 2

Worked Reasoning. Set $x=-1$ after multiplying by $(x+1)(x+2)$. This gives $2=A(1)$, so $A=2$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 2.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 2.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 2.

Trick / Tip. Question 28 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 2, not just look numerically close.

Q029

Mathematics Quadratics and Algebra

MCQ Focus. The partial fraction decomposition of $5/(x^2-1)$ contains coefficient of $1/(x-1)$ equal to:

- A. $\frac{5}{2}$
- B. $-\frac{5}{2}$
- C. 5
- D. -5

Fast Route. Use Vieta, discriminant, substitution, or cover-up before attempting a long solution.

Trap Watch. Sign errors and confusing sum/product of roots create most wrong options.

Answer. A $\frac{5}{2}$

Worked Reasoning. Since $x^2 - 1 = (x-1)(x+1)$, the coefficient of $1/(x-1)$ is obtained by $x=1$: $5/(1+1) = \frac{5}{2}$.

Why the other options are wrong.

- ◇ Option B: comes from a sign slip in the coefficient or discriminant step; it leads away from $\frac{5}{2}$.
- ◇ Option C: stops before the algebraic simplification that the options require; it leads away from $\frac{5}{2}$.
- ◇ Option D: uses the sum of roots where the product or substitution condition is being tested; it leads away from $\frac{5}{2}$.

Trick / Tip. The useful shortcut here is tied to complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\frac{5}{2}$, not just look numerically close.



Q030

Mathematics Complex Numbers

MCQ Focus. If $x/(x^2-4)=A/(x-2)+B/(x+2)$, then $A+B$ is:

- A. 0
- B. 1
- C. -1
- D. 2

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 1

Worked Reasoning. By cover-up, $A=\frac{1}{2}$ and $B=\frac{1}{2}$, so $A+B=1$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 1.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 1.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 1.

Trick / Tip. For this Complex Numbers pattern, notice complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 1, not just look numerically close.



Q031

Mathematics Complex Numbers

MCQ Focus. The 10th term of AP 7, 11, 15, ... is:

- A. 39
- B. 43
- C. 47
- D. 51

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 43

Worked Reasoning. In an AP, $a_n = a + (n-1)d$. Here $a=7$, $d=4$, so $a_{10} = 7 + 9(4) = 43$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 43.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 43.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 43.

Trick / Tip. This question rewards recognising arithmetic progression: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with 43, not just look numerically close.

Q032

Mathematics Complex Numbers

MCQ Focus. If the 4th and 9th terms of an AP are 13 and 28, common difference is:

- A. 2
- B. 3
- C. 4
- D. 5

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 3

Worked Reasoning. $a_9 - a_4 = 5d = 28 - 13 = 15$, so $d = 3$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 3.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 3.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 3.

Trick / Tip. Question 32 is a arithmetic progression item: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with 3, not just look numerically close.



Q033

Mathematics Complex Numbers

MCQ Focus. The sum of first 20 odd positive integers is:

- A. 200
- B. 300
- C. 400
- D. 420

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 400

Worked Reasoning. The sum of the first n odd positive integers is n^2 . For $n=20$, the sum is 400.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 400.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 400.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 400.

Trick / Tip. The useful shortcut here is tied to complex numbers: For complex-number MCQs, compute only the part asked for; if the question asks the real part, do not waste time finishing the imaginary part.. The correct option should agree with 400, not just look numerically close.



Q034

Mathematics Complex Numbers

MCQ Focus. For a GP, $a=3$ and $r=2$. The sum of first 6 terms is:

- A. 93
- B. 189
- C. 192
- D. 96

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 189

Worked Reasoning. Use $S_n = a(r^n - 1)/(r - 1)$. $S_6 = 3(2^6 - 1) = 189$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 189.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 189.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 189.

Trick / Tip. For this Complex Numbers pattern, notice geometric progression: use the common ratio structure; for a positive middle term, choose the positive geometric mean. The correct option should agree with 189, not just look numerically close.



Q035

Mathematics Complex Numbers

MCQ Focus. If 2, x , 18 are in GP and x is positive, x is:

- A. 4
- B. 6
- C. 9
- D. 12

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 6

Worked Reasoning. For three GP terms, $x^2=2(18)=36$. Since x is positive, $x=6$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 6.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 6.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 6.

Trick / Tip. This question rewards recognising geometric progression: use the common ratio structure; for a positive middle term, choose the positive geometric mean. The correct option should agree with 6, not just look numerically close.

Q036

Mathematics Complex Numbers

MCQ Focus. If arithmetic mean of two positive numbers is 10 and geometric mean is 8, their product is:

- A. 16
- B. 64
- C. 80
- D. 100

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 64

Worked Reasoning. The geometric mean is $\sqrt{ab}$. If $GM=8$, then $ab=64$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 64.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 64.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 64.

Trick / Tip. Question 36 is a complex numbers item: For complex-number MCQs, compute only the part asked for; if the question asks the real part, do not waste time finishing the imaginary part.. The correct option should agree with 64, not just look numerically close.



Q037

Mathematics Mathematics Pattern Recognition

MCQ Focus. How many 4-digit numbers can be formed from digits 1,2,3,4,5 without repetition?

- A. 20
- B. 60
- C. 120
- D. 625

Fast Route. Spot the tested rule first and avoid long working.

Trap Watch. The distractor usually uses the right topic but the wrong rule.

Answer. C 120

Worked Reasoning. Five choices are available for the first digit, then 4, 3, and 2.
Total = $5 \cdot 4 \cdot 3 \cdot 2 = 120$.

Why the other options are wrong.

- ◇ Option A: does not follow the rule used in the worked solution; it leads away from 120.
- ◇ Option B: uses a nearby but wrong pattern; it leads away from 120.
- ◇ Option D: does not follow the rule used in the worked solution; it leads away from 120.

Trick / Tip. The useful shortcut here is tied to mathematics pattern recognition: Use the tested rule for mathematics pattern recognition, then compare against the options.. The correct option should agree with 120, not just look numerically close.

Q038

Mathematics Complex Numbers

MCQ Focus. Number of ways to choose 3 students from 8 is:

- A. 24
- B. 56
- C. 336
- D. 512

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 56

Worked Reasoning. Use combinations: $C(8, 3) = 56$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 56.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 56.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 56.

Trick / Tip. For this Complex Numbers pattern, notice combination count: decide whether order matters. If it does not, use combinations rather than permutations. The correct option should agree with 56, not just look numerically close.



Q039

Mathematics Complex Numbers

MCQ Focus. If two dice are rolled, probability of getting sum 9 is:

- A. $\frac{1}{12}$
- B. $\frac{1}{9}$
- C. $\frac{1}{6}$
- D. $\frac{1}{4}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** $\frac{1}{9}$

Worked Reasoning. Sum 9 occurs in 4 outcomes: (3,6),(4,5),(5,4),(6,3).
Probability = $\frac{4}{36} = \frac{1}{9}$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{1}{9}$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\frac{1}{9}$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{1}{9}$.

Trick / Tip. This question rewards recognising complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\frac{1}{9}$, not just look numerically close.

Q040

Mathematics Complex Numbers

MCQ Focus. A card is drawn from a standard deck. Probability that it is a king or a heart is:

- A. $\frac{4}{13}$
- B. $\frac{5}{13}$
- C. $\frac{17}{52}$
- D. $\frac{1}{2}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $\frac{4}{13}$

Worked Reasoning. Kings or hearts: $4 + 13 - 1 = 16$ favorable cards.
Probability = $\frac{16}{52} = \frac{4}{13}$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\frac{4}{13}$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\frac{4}{13}$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{4}{13}$.

Trick / Tip. Question 40 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\frac{4}{13}$, not just look numerically close.



Q041

Mathematics Complex Numbers

MCQ Focus. Number of arrangements of letters of the word LEVEL is:

- A. 20
- B. 30
- C. 60
- D. 120

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 30

Worked Reasoning. LEVEL has 5 letters with two Ls and two Es. Arrangements = $5!/(2!2!)=30$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 30.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 30.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 30.

Trick / Tip. The useful shortcut here is tied to complex numbers: When ω appears, immediately write $1+\omega+\omega^2=0$ and $\omega^3=1$ before touching the options.. The correct option should agree with 30, not just look numerically close.

Q042

Mathematics Complex Numbers

MCQ Focus. If $P(A) = 0.6$, $P(B) = 0.5$ and $P(A \cap B) = 0.2$, then $P(A \cup B)$ is:

- A. 0.7
- B. 0.8
- C. 0.9
- D. 1.1

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 0.9

Worked Reasoning. Use $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - 0.2 = 0.9$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 0.9.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 0.9.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 0.9.

Trick / Tip. For this Complex Numbers pattern, notice set inclusion-exclusion: use $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ and solve for the missing region. The correct option should agree with 0.9, not just look numerically close.



Q043

Mathematics Complex Numbers

MCQ Focus. In how many ways can 5 boys and 3 girls sit in a row if all girls sit together?

- A. 720
- B. 1440
- C. 2880
- D. 4320

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. D 4320

Worked Reasoning. Treat the 3 girls as one block. Then 6 units can be arranged in $6!$ ways and girls within the block in $3!$ ways: $6! \cdot 3! = 4320$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 4320.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 4320.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 4320.

Trick / Tip. This question rewards recognising complex numbers: Use conjugates for division and the four-step cycle for powers of i ; these two moves solve most complex-number distractors.. The correct option should agree with 4320, not just look numerically close.



Q044

Mathematics Complex Numbers

MCQ Focus. If a coin is tossed 5 times, probability of exactly 3 heads is:

- A. $\frac{5}{16}$
- B. $\frac{3}{16}$
- C. $\frac{10}{32}$
- D. $\frac{1}{2}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. ☒ C $\frac{10}{32}$

Worked Reasoning. Exactly 3 heads in 5 tosses has probability $C(5,3)/2^5 = \frac{10}{32}$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{10}{32}$.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\frac{10}{32}$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{10}{32}$.

Trick / Tip. Question 44 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\frac{10}{32}$, not just look numerically close.



Q045

Mathematics Complex Numbers

MCQ Focus. The coefficient of x^3 in $(1 + x)^8$ is:

- A. 28
- B. 35
- C. 56
- D. 70

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 56

Worked Reasoning. The coefficient of x^3 in $(1 + x)^8$ is $C(8, 3) = 56$.

Why the other options are wrong.

- ◊ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 56.
- ◊ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 56.
- ◊ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 56.

Trick / Tip. The useful shortcut here is tied to complex numbers: For complex-number MCQs, compute only the part asked for; if the question asks the real part, do not waste time finishing the imaginary part.. The correct option should agree with 56, not just look numerically close.



Q046

Mathematics

Complex Numbers

MCQ Focus. The middle term in expansion of $(x + a)^6$ is:

- A. $15x^3a^3$
- B. $20x^3a^3$
- C. $15x^4a^2$
- D. $6x^5a$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B $20x^3a^3$

Worked Reasoning. For $n=6$, the middle term is the 4th term:
 $C(6,3)x^3a^3=20x^3a^3$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $20x^3a^3$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $20x^3a^3$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $20x^3a^3$.

Trick / Tip. For this Complex Numbers pattern, notice complex numbers: Use conjugates for division and the four-step cycle for powers of i ; these two moves solve most complex-number distractors.. The correct option should agree with $20x^3a^3$, not just look numerically close.



Q047

Mathematics Complex Numbers

MCQ Focus. The sum of coefficients in $(2x - 1)^7$ is:

- A. 0
- B. 1
- C. -1
- D. 128

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 1

Worked Reasoning. The sum of coefficients is found by putting $x=1$: $(2(1)-1)^7=1$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 1.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 1.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 1.

Trick / Tip. This question rewards recognising complex numbers: When ω appears, immediately write $1+\omega+\omega^2=0$ and $\omega^3=1$ before touching the options.. The correct option should agree with 1, not just look numerically close.

Q048

Mathematics Complex Numbers

MCQ Focus. The coefficient of x^2 in $(1 - 2x)^5$ is:

- A. 10
- B. 20
- C. 40
- D. -20

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 40

Worked Reasoning. The x^2 coefficient is $C(5,2)(-2)^2 = 10 \cdot 4 = 40$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 40.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 40.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 40.

Trick / Tip. Question 48 is a quadratic structure item: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with 40, not just look numerically close.



Q049

Mathematics Complex Numbers

MCQ Focus. In expansion of $\left(x + \frac{1}{x}\right)^6$, the constant term is:

- A. 10
- B. 15
- C. 20
- D. 30

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 20

Worked Reasoning. The constant term occurs when powers balance: $C(6, 3) = 20$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 20.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 20.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 20.

Trick / Tip. The useful shortcut here is tied to complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 20, not just look numerically close.



Q050

Mathematics Complex Numbers

MCQ Focus. If $\sin \theta = \frac{3}{5}$ and θ is acute, $\cos \theta$ is:

- A. $\frac{2}{5}$
- B. $\frac{3}{5}$
- C. $\frac{4}{5}$
- D. $\frac{5}{4}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C $\frac{4}{5}$

Worked Reasoning. Use $\sin^2 \theta + \cos^2 \theta = 1$. With $\sin \theta = \frac{3}{5}$ and θ acute, $\cos \theta = \frac{4}{5}$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{4}{5}$.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\frac{4}{5}$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{4}{5}$.

Trick / Tip. For this Complex Numbers pattern, notice complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\frac{4}{5}$, not just look numerically close.



Q051

Mathematics Complex Numbers

MCQ Focus. The period of $\tan(2x)$ is:

- A. π
- B. $\pi/2$
- C. 2π
- D. 4π

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** $\pi/2$

Worked Reasoning. The period of $\tan(kx)$ is π/k . For $k=2$, the period is $\pi/2$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\pi/2$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\pi/2$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\pi/2$.

Trick / Tip. This question rewards recognising complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\pi/2$, not just look numerically close.

Q052

Mathematics Complex Numbers

MCQ Focus. If $\sin A = \cos A$ and A is acute, then A is:

- A. 30°
- B. 45°
- C. 60°
- D. 90°

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 45°

Worked Reasoning. For an acute angle, $\sin A = \cos A$ only at $A = 45^\circ$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 45° .
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 45° .
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 45° .

Trick / Tip. Question 52 is a trigonometric identity item: search for a sum-difference, double-angle, or reciprocal identity before substituting values. The correct option should agree with 45° , not just look numerically close.



Q053

Mathematics Complex Numbers

MCQ Focus. The value of $\sin 75^\circ \cos 15^\circ - \cos 75^\circ \sin 15^\circ$ is:

- A. 0
- B. $\frac{1}{2}$
- C. $\sqrt{3}/2$
- D. 1

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C $\sqrt{3}/2$

Worked Reasoning. Use $\sin a \cos b - \cos a \sin b = \sin(a - b)$. This equals $\sin 60^\circ = \sqrt{3}/2$.

Why the other options are wrong.

- ◊ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\sqrt{3}/2$.
- ◊ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\sqrt{3}/2$.
- ◊ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\sqrt{3}/2$.

Trick / Tip. The useful shortcut here is tied to complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\sqrt{3}/2$, not just look numerically close.



Q054

Mathematics Complex Numbers

MCQ Focus. If $\tan \theta = \frac{3}{4}$, then $\sin \theta$ for acute θ is:

- A. $\frac{3}{5}$
- B. $\frac{4}{5}$
- C. $\frac{5}{3}$
- D. $\frac{5}{4}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $\frac{3}{5}$

Worked Reasoning. For $\tan \theta = \frac{3}{4}$, take opposite=3 and adjacent=4, so hypotenuse=5 and $\sin \theta = \frac{3}{5}$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\frac{3}{5}$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\frac{3}{5}$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{3}{5}$.

Trick / Tip. For this Complex Numbers pattern, notice complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\frac{3}{5}$, not just look numerically close.



Q055

Mathematics Complex Numbers

MCQ Focus. The maximum value of $3\sin x + 4\cos x$ is:

- A. 4
- B. 5
- C. 7
- D. 12

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 5

Worked Reasoning. The maximum of $a \sin x + b \cos x$ is $\sqrt{a^2 + b^2}$. Here $\sqrt{3^2 + 4^2} = 5$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 5.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 5.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 5.

Trick / Tip. This question rewards recognising trigonometric identity: search for a sum-difference, double-angle, or reciprocal identity before substituting values. The correct option should agree with 5, not just look numerically close.

Q056

Mathematics Complex Numbers

MCQ Focus. If $\sec \theta = \frac{5}{4}$ and θ is acute, $\tan \theta$ is:

- A. $\frac{3}{4}$
- B. $\frac{4}{3}$
- C. $\frac{5}{3}$
- D. $\frac{5}{4}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $\frac{3}{4}$

Worked Reasoning. $\sec \theta = \frac{5}{4}$ means $\cos \theta = \frac{4}{5}$. The 3-4-5 triangle gives $\tan \theta = \frac{3}{4}$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\frac{3}{4}$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\frac{3}{4}$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{3}{4}$.

Trick / Tip. Question 56 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\frac{3}{4}$, not just look numerically close.



Q057

Mathematics Trigonometry

MCQ Focus. $\sin 2A$ equals:

- A. $2\sin A \cos A$
- B. $\sin^2 A - \cos^2 A$
- C. $1 - 2\sin^2 A$
- D. $\tan A/2$

Fast Route. Match the expression to a standard identity or special-angle value before calculating.

Trap Watch. Do not mix identities; a similar-looking identity can give a listed but wrong option.

Answer. A $2\sin A \cos A$

Worked Reasoning. This is the standard double-angle identity: $\sin 2A = 2 \sin A \cos A$.

Why the other options are wrong.

- ◇ Option B: mixes special-angle values or forgets the angle condition; it leads away from $2\sin A \cos A$.
- ◇ Option C: drops a square, double-angle, or reciprocal relationship; it leads away from $2\sin A \cos A$.
- ◇ Option D: uses a similar-looking identity that does not match the expression; it leads away from $2\sin A \cos A$.

Trick / Tip. The useful shortcut here is tied to complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $2\sin A \cos A$, not just look numerically close.

Q058

Mathematics Complex Numbers

MCQ Focus. If $\cos 2A = \frac{1}{2}$ and A is acute, A is:

- A. 15°
- B. 30°
- C. 45°
- D. 60°

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 30°

Worked Reasoning. If $\cos 2A = \frac{1}{2}$, then $2A = 60^\circ$ for acute A , so $A = 30^\circ$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 30° .
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 30° .
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 30° .

Trick / Tip. For this Complex Numbers pattern, notice complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 30° , not just look numerically close.



Q059

Mathematics Complex Numbers

MCQ Focus. The value of $\cot 45^\circ + \tan 45^\circ$ is:

- A. 0
- B. 1
- C. 2
- D. $\sqrt{2}$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 2

Worked Reasoning. $\cot 45^\circ = 1$ and $\tan 45^\circ = 1$, so the sum is 2.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 2.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 2.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 2.

Trick / Tip. This question rewards recognising trigonometric ratio: use the triangle ratio implied by the given function, then derive the requested function. The correct option should agree with 2, not just look numerically close.

Q060

Mathematics Complex Numbers

MCQ Focus. If $\sin x = 0$ in $0 \leq x \leq 2\pi$, number of solutions is:

- A. 1
- B. 2
- C. 3
- D. 4

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 3

Worked Reasoning. In $0 \leq x \leq 2\pi$, $\sin x = 0$ at $x = 0, \pi$, and 2π : three solutions.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 3.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 3.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 3.

Trick / Tip. Question 60 is a trigonometric identity item: search for a sum-difference, double-angle, or reciprocal identity before substituting values. The correct option should agree with 3, not just look numerically close.



Q061

Mathematics Complex Numbers

MCQ Focus. The value of $\cos^2 x - \sin^2 x$ is:

- A. $\sin 2x$
- B. $\cos 2x$
- C. $\tan 2x$
- D. 1

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B $\cos 2x$

Worked Reasoning. $\cos^2 x - \sin^2 x$ is the double-angle identity for $\cos 2x$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\cos 2x$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\cos 2x$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\cos 2x$.

Trick / Tip. The useful shortcut here is tied to trigonometric identity: search for a sum-difference, double-angle, or reciprocal identity before substituting values. The correct option should agree with $\cos 2x$, not just look numerically close.



Q062

Mathematics Complex Numbers

MCQ Focus. If $\tan A + \cot A = 2$, then A is:

- A. 30°
- B. 45°
- C. 60°
- D. 90°

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 45°

Worked Reasoning. $\tan A + \cot A \geq 2$, with equality when $\tan A = 1$. Thus $A = 45^\circ$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 45° .
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 45° .
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 45° .

Trick / Tip. For this Complex Numbers pattern, notice trigonometric ratio: use the triangle ratio implied by the given function, then derive the requested function. The correct option should agree with 45° , not just look numerically close.



Q063

Mathematics Quadratics and Algebra

MCQ Focus. $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2}$ equals:

- A. 0
- B. 2
- C. 4
- D. Does not exist

Fast Route. Use Vieta, discriminant, substitution, or cover-up before attempting a long solution.

Trap Watch. Sign errors and confusing sum/product of roots create most wrong options.

Answer. C 4

Worked Reasoning. Factor $x^2 - 4 = (x-2)(x+2)$, cancel $x-2$, then substitute $x=2$ to get 4.

Why the other options are wrong.

- ◇ Option A: uses the sum of roots where the product or substitution condition is being tested; it leads away from 4.
- ◇ Option B: comes from a sign slip in the coefficient or discriminant step; it leads away from 4.
- ◇ Option D: uses the sum of roots where the product or substitution condition is being tested; it leads away from 4.

Trick / Tip. This question rewards recognising complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 4, not just look numerically close.

Q064

Mathematics Trigonometry

MCQ Focus. $\lim_{x \rightarrow 0} \sin(5x)/x$ equals:

- A. 0
- B. 1
- C. 5
- D. $\frac{1}{5}$

Fast Route. Match the expression to a standard identity or special-angle value before calculating.

Trap Watch. Do not mix identities; a similar-looking identity can give a listed but wrong option.

Answer. **C** 5

Worked Reasoning. Use the standard limit $\sin \frac{u}{u} = 1$. $\sin(5x)/x = 5[\sin(5x)/(5x)]$, so the limit is 5.

Why the other options are wrong.

- ◇ Option A: uses a similar-looking identity that does not match the expression; it leads away from 5.
- ◇ Option B: mixes special-angle values or forgets the angle condition; it leads away from 5.
- ◇ Option D: uses a similar-looking identity that does not match the expression; it leads away from 5.

Trick / Tip. Question 64 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 5, not just look numerically close.

Q065

Mathematics Complex Numbers

MCQ Focus. If $f(x) = x^3 - 2x$, then $f'(2)$ is:

- A. 8
- B. 10
- C. 12
- D. 16

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 10

Worked Reasoning. $f'(x) = 3x^2 - 2$, so $f'(2) = 12 - 2 = 10$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 10.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 10.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 10.

Trick / Tip. The useful shortcut here is tied to function rule: substitute carefully and respect the order of composition or inversion. The correct option should agree with 10, not just look numerically close.

Q066

Mathematics Complex Numbers

MCQ Focus. Derivative of $\ln(x^2 + 1)$ is:

- A. $1/(x^2+1)$
- B. $2x/(x^2+1)$
- C. $2/(x^2+1)$
- D. $x/(x^2+1)$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** $2x/(x^2+1)$

Worked Reasoning. By the chain rule, $d/dx \ln(x^2 + 1) = 2x/(x^2 + 1)$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $2x/(x^2+1)$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $2x/(x^2+1)$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $2x/(x^2+1)$.

Trick / Tip. For this Complex Numbers pattern, notice complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $2x/(x^2+1)$, not just look numerically close.



Q067

Mathematics Complex Numbers

MCQ Focus. If $y=e^{3x}$, dy/dx is:

- A. e^{3x}
- B. $3e^{3x}$
- C. $x e^{3x}$
- D. e^x

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** $3e^{3x}$

Worked Reasoning. By the chain rule, $d/dx e^{3x}=3e^{3x}$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $3e^{3x}$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $3e^{3x}$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $3e^{3x}$.

Trick / Tip. This question rewards recognising complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $3e^{3x}$, not just look numerically close.

Q068

Mathematics Complex Numbers

MCQ Focus. Derivative of $x^2 \sin x$ is:

- A. $2x \sin x + x^2 \cos x$
- B. $2x \cos x$
- C. $x^2 \cos x$
- D. $2x \sin x - x^2 \cos x$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $2x \sin x + x^2 \cos x$

Worked Reasoning. Use the product rule: derivative of $x^2 \sin x$ is $2x \sin x + x^2 \cos x$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $2x \sin x + x^2 \cos x$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $2x \sin x + x^2 \cos x$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $2x \sin x + x^2 \cos x$.

Trick / Tip. Question 68 is a quadratic structure item: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with $2x \sin x + x^2 \cos x$, not just look numerically close.



Q069

Mathematics Complex Numbers

MCQ Focus. If $y = \sin^2 x$, dy/dx is:

- A. $\sin 2x$
- B. $\cos 2x$
- C. $2\sin x$
- D. $2\cos x$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $\sin 2x$

Worked Reasoning. $d/dx \sin^2 x = 2 \sin x \cos x = \sin 2x$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\sin 2x$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\sin 2x$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\sin 2x$.

Trick / Tip. The useful shortcut here is tied to complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\sin 2x$, not just look numerically close.

Q070

Mathematics Complex Numbers

MCQ Focus. The slope of $y=x^2 - 4x+1$ at $x=3$ is:

- A. 0
- B. 1
- C. 2
- D. 6

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 2

Worked Reasoning. $\frac{dy}{dx} = 2x - 4$. At $x=3$, slope=2.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 2.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 2.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 2.

Trick / Tip. For this Complex Numbers pattern, notice quadratic structure: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with 2, not just look numerically close.



Q071

Mathematics Complex Numbers

MCQ Focus. Integral of $2x \cos(x^2) dx$ is:

- A. $\sin(x^2) + C$
- B. $\cos(x^2) + C$
- C. $x^2 \sin x + C$
- D. $2\sin(x^2) + C$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $\sin(x^2) + C$

Worked Reasoning. Let $u = x^2$, so $du = 2x dx$. The integral becomes $\sin(x^2) + C$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\sin(x^2) + C$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\sin(x^2) + C$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\sin(x^2) + C$.

Trick / Tip. This question rewards recognising quadratic structure: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with $\sin(x^2) + C$, not just look numerically close.

Q072

Mathematics Complex Numbers

MCQ Focus. Integral of $1/(x \ln x)$ dx is:

- A. $\ln x + C$
- B. $\ln(\ln x) + C$
- C. $1/\ln x + C$
- D. $x \ln x + C$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** $\ln(\ln x) + C$

Worked Reasoning. Let $u = \ln x$, so $du = dx/x$. The integral is $\ln(\ln x) + C$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\ln(\ln x) + C$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\ln(\ln x) + C$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\ln(\ln x) + C$.

Trick / Tip. Question 72 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\ln(\ln x) + C$, not just look numerically close.



Q073

Mathematics Complex Numbers

MCQ Focus. Integral of e^{2x} dx is:

- A. $e^{2x} + C$
- B. $2e^{2x} + C$
- C. $(\frac{1}{2})e^{2x} + C$
- D. $x e^{2x} + C$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C $(\frac{1}{2})e^{2x} + C$

Worked Reasoning. Integral of e^{2x} is $(\frac{1}{2})e^{2x} + C$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $(\frac{1}{2})e^{2x} + C$.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $(\frac{1}{2})e^{2x} + C$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $(\frac{1}{2})e^{2x} + C$.

Trick / Tip. The useful shortcut here is tied to complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $(\frac{1}{2})e^{2x} + C$, not just look numerically close.

Q074

Mathematics Quadratics and Algebra

MCQ Focus. Integral from 0 to 1 of $3x^2$ dx equals:

- A. 0
- B. 1
- C. 2
- D. 3

Fast Route. Use Vieta, discriminant, substitution, or cover-up before attempting a long solution.

Trap Watch. Sign errors and confusing sum/product of roots create most wrong options.

Answer. B 1

Worked Reasoning. Integral of $3x^2$ from 0 to 1 is $[x^3]_0^1=1$.

Why the other options are wrong.

- ◇ Option A: uses the sum of roots where the product or substitution condition is being tested; it leads away from 1.
- ◇ Option C: stops before the algebraic simplification that the options require; it leads away from 1.
- ◇ Option D: uses the sum of roots where the product or substitution condition is being tested; it leads away from 1.

Trick / Tip. For this Quadratics and Algebra pattern, notice quadratic structure: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with 1, not just look numerically close.



Q075

Mathematics Complex Numbers

MCQ Focus. The area under $y=2x$ from $x=1$ to $x=3$ is:

- A. 4
- B. 6
- C. 8
- D. 10

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 8

Worked Reasoning. Area is integral of $2x$ from 1 to 3: $[x^2]_1^3 = 9 - 1 = 8$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 8.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 8.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 8.

Trick / Tip. This question rewards recognising complex numbers: For complex-number MCQs, compute only the part asked for; if the question asks the real part, do not waste time finishing the imaginary part.. The correct option should agree with 8, not just look numerically close.

Q076

Mathematics Complex Numbers

MCQ Focus. If $\frac{dy}{dx} = 6x$ and $y(0) = 4$, then y is:

- A. $3x^2+4$
- B. $6x^2+4$
- C. $3x+4$
- D. x^3+4

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $3x^2+4$

Worked Reasoning. Integrate $\frac{dy}{dx} = 6x$ to get $y=3x^2+C$. Since $y(0) = 4$, $C=4$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $3x^2+4$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $3x^2+4$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $3x^2+4$.

Trick / Tip. Question 76 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $3x^2+4$, not just look numerically close.



Q077

Mathematics Complex Numbers

MCQ Focus. The equation of the line through (2,3) with slope -1 is:

- A. $x + y = 5$
- B. $x - y = 5$
- C. $y = x + 1$
- D. $x + y = 1$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $x + y = 5$

Worked Reasoning. Use $y - 3 = -1(x - 2)$, which simplifies to $x + y = 5$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $x + y = 5$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $x + y = 5$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $x + y = 5$.

Trick / Tip. The useful shortcut here is tied to complex numbers: When ω appears, immediately write $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$ before touching the options.. The correct option should agree with $x + y = 5$, not just look numerically close.



Q078

Mathematics Complex Numbers

MCQ Focus. Distance between (1,2) and (4,6) is:

- A. 3
- B. 4
- C. 5
- D. 7

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 5

Worked Reasoning. Distance = $\sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = 5$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 5.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 5.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 5.

Trick / Tip. For this Complex Numbers pattern, notice trigonometric ratio: use the triangle ratio implied by the given function, then derive the requested function. The correct option should agree with 5, not just look numerically close.



Q079

Mathematics Complex Numbers

MCQ Focus. Midpoint of $(-2,5)$ and $(4,-1)$ is:

- A. $(1,2)$
- B. $(2,1)$
- C. $(-1,2)$
- D. $(1,-2)$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. A $(1,2)$

Worked Reasoning. Midpoint $= ((-2+4)/2, (5-1)/2) = (1,2)$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $(1,2)$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $(1,2)$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $(1,2)$.

Trick / Tip. This question rewards recognising complex numbers: Use conjugates for division and the four-step cycle for powers of i ; these two moves solve most complex-number distractors.. The correct option should agree with $(1,2)$, not just look numerically close.

Q080

Mathematics Quadratics and Algebra

MCQ Focus. Circle $x^2 + y^2 - 6x + 8y = 0$ has radius:

- A. 3
- B. 4
- C. 5
- D. 10

Fast Route. Use Vieta, discriminant, substitution, or cover-up before attempting a long solution.

Trap Watch. Sign errors and confusing sum/product of roots create most wrong options.

Answer. C 5

Worked Reasoning. Complete squares: $(x - 3)^2 + (y + 4)^2 = 25$, so radius=5.

Why the other options are wrong.

- ◇ Option A: uses the sum of roots where the product or substitution condition is being tested; it leads away from 5.
- ◇ Option B: comes from a sign slip in the coefficient or discriminant step; it leads away from 5.
- ◇ Option D: uses the sum of roots where the product or substitution condition is being tested; it leads away from 5.

Trick / Tip. Question 80 is a quadratic structure item: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with 5, not just look numerically close.



Q081

Mathematics Complex Numbers

MCQ Focus. The center of $x^2 + y^2 + 4x - 10y + 20 = 0$ is:

- A. (2,-5)
- B. (-2,5)
- C. (4,-10)
- D. (-4,10)

Fast Route. Use the circle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B (-2,5)

Worked Reasoning. Complete squares: $(x + 2)^2 + (y - 5)^2 = 9$, so center = (-2,5).

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from (-2,5).
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from (-2,5).
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from (-2,5).

Trick / Tip. The useful shortcut here is tied to quadratic structure: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with (-2,5), not just look numerically close.

Q082

Mathematics

Mathematics Pattern Recognition

MCQ Focus. The conic $y^2=12x$ has focus:

- A. (3,0)
- B. (-3,0)
- C. (0,3)
- D. (0,-3)

Fast Route. Spot the tested rule first and avoid long working.

Trap Watch. The distractor usually uses the right topic but the wrong rule.

Answer. A (3,0)

Worked Reasoning. For $y^2=4ax$, $4a=12$, so $a=3$ and focus=(3,0).

Why the other options are wrong.

- ◇ Option B: uses a nearby but wrong pattern; it leads away from (3,0).
- ◇ Option C: conflicts with the calculated result; it leads away from (3,0).
- ◇ Option D: does not follow the rule used in the worked solution; it leads away from (3,0).

Trick / Tip. For this Mathematics Pattern Recognition pattern, notice mathematics pattern recognition: Use the tested rule for mathematics pattern recognition, then compare against the options.. The correct option should agree with (3,0), not just look numerically close.



Q083

Mathematics Complex Numbers

MCQ Focus. The eccentricity of a parabola is:

- A. 0
- B. 1
- C. Between 0 and 1
- D. Greater than 1

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 1

Worked Reasoning. The eccentricity of every parabola is 1.

Why the other options are wrong.

- ◊ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 1.
- ◊ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 1.
- ◊ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 1.

Trick / Tip. This question rewards recognising complex numbers: When ω appears, immediately write $1+\omega+\omega^2=0$ and $\omega^3=1$ before touching the options.. The correct option should agree with 1, not just look numerically close.

Q084

Mathematics Quadratics and Algebra

MCQ Focus. The ellipse $x^{\frac{2}{25}} + y^{\frac{2}{9}} = 1$ has major axis length:

- A. 3
- B. 5
- C. 6
- D. 10

Fast Route. Use Vieta, discriminant, substitution, or cover-up before attempting a long solution.

Trap Watch. Sign errors and confusing sum/product of roots create most wrong options.

Answer. D 10

Worked Reasoning. For $x^{\frac{2}{25}} + y^{\frac{2}{9}} = 1$, $a=5$, so major axis length $=2a=10$.

Why the other options are wrong.

- ◇ Option A: uses the sum of roots where the product or substitution condition is being tested; it leads away from 10.
- ◇ Option B: comes from a sign slip in the coefficient or discriminant step; it leads away from 10.
- ◇ Option C: stops before the algebraic simplification that the options require; it leads away from 10.

Trick / Tip. Question 84 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 10, not just look numerically close.



Q085

Mathematics Quadratics and Algebra

MCQ Focus. The hyperbola $x^{\frac{2}{9}} - y^{\frac{2}{16}} = 1$ has transverse axis length:

- A. 3
- B. 6
- C. 8
- D. 18

Fast Route. Use Vieta, discriminant, substitution, or cover-up before attempting a long solution.

Trap Watch. Sign errors and confusing sum/product of roots create most wrong options.

Answer. B 6

Worked Reasoning. For $x^{\frac{2}{9}} - y^{\frac{2}{16}} = 1$, $a=3$, so transverse axis length $= 2a=6$.

Why the other options are wrong.

- ◇ Option A: uses the sum of roots where the product or substitution condition is being tested; it leads away from 6.
- ◇ Option C: stops before the algebraic simplification that the options require; it leads away from 6.
- ◇ Option D: uses the sum of roots where the product or substitution condition is being tested; it leads away from 6.

Trick / Tip. The useful shortcut here is tied to complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 6, not just look numerically close.



Q086

Mathematics Coordinate Geometry and Vectors

MCQ Focus. If vectors $a=(2, -1, 3)$ and $b=(1, 4, -2)$, then $a \cdot b$ equals:

- A. -8
- B. -4
- C. 0
- D. 4

Fast Route. Name the formula form first: distance, midpoint, line, conic, magnitude, dot, or cross product.

Trap Watch. A swapped coordinate or wrong axis length can still look numerically reasonable.

Answer. A -8

Worked Reasoning. Dot product: $2(1)+(-1)(4)+3(-2)=2-4-6=-8$.

Why the other options are wrong.

- ◇ Option B: confuses axis length, radius, focus, magnitude, dot product, or cross product; it leads away from -8.
- ◇ Option C: uses a plausible number from the data but not the required geometric quantity; it leads away from -8.
- ◇ Option D: uses the wrong coordinate formula or swaps the coordinate roles; it leads away from -8.

Trick / Tip. For this Coordinate Geometry and Vectors pattern, notice vector operation: decide whether the question asks for dot product, cross product, or magnitude before using components. The correct option should agree with -8, not just look numerically close.



Q087

Mathematics Complex Numbers

MCQ Focus. Magnitude of vector $2i - 3j + 6k$ is:

- A. 5
- B. 6
- C. 7
- D. 49

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 7

Worked Reasoning. $\text{Magnitude} = \sqrt{2^2 + (-3)^2 + 6^2} = \sqrt{49} = 7$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 7.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 7.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 7.

Trick / Tip. This question rewards recognising vector operation: decide whether the question asks for dot product, cross product, or magnitude before using components. The correct option should agree with 7, not just look numerically close.

Q088

Mathematics Coordinate Geometry and Vectors

MCQ Focus. If a and b are perpendicular non-zero vectors, then:

- A. $a \cdot b = 0$
- B. $a \times b = 0$
- C. $|a| = |b|$
- D. $a + b = 0$

Fast Route. Name the formula form first: distance, midpoint, line, conic, magnitude, dot, or cross product.

Trap Watch. A swapped coordinate or wrong axis length can still look numerically reasonable.

Answer. **A** $a \cdot b = 0$

Worked Reasoning. Perpendicular non-zero vectors have dot product zero.

Why the other options are wrong.

- ◇ Option B: confuses axis length, radius, focus, magnitude, dot product, or cross product; it leads away from $a \cdot b = 0$.
- ◇ Option C: uses a plausible number from the data but not the required geometric quantity; it leads away from $a \cdot b = 0$.
- ◇ Option D: uses the wrong coordinate formula or swaps the coordinate roles; it leads away from $a \cdot b = 0$.

Trick / Tip. Question 88 is a vector operation item: decide whether the question asks for dot product, cross product, or magnitude before using components. The correct option should agree with $a \cdot b = 0$, not just look numerically close.

Q089

Mathematics Complex Numbers

MCQ Focus. The value of $i \times j$ is:

- A. i
- B. j
- C. k
- D. $-k$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C k

Worked Reasoning. The standard unit-vector relation is $i \times j = k$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from k .
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from k .
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from k .

Trick / Tip. The useful shortcut here is tied to complex numbers: When ω appears, immediately write $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$ before touching the options.. The correct option should agree with k , not just look numerically close.

Q090

Mathematics Complex Numbers

MCQ Focus. For maximizing $Z=3x+2y$ subject to $x+y \leq 4$, $x \leq 3$, $y \leq 3$, $x,y \geq 0$, the maximum Z is:

- A. 8
- B. 10
- C. 11
- D. 12

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 11

Worked Reasoning. Check corner points. At $(3,1)$, $Z=3(3)+2(1)=11$, the maximum.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 11.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 11.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 11.

Trick / Tip. For this Complex Numbers pattern, notice complex numbers: For complex-number MCQs, compute only the part asked for; if the question asks the real part, do not waste time finishing the imaginary part.. The correct option should agree with 11, not just look numerically close.



Q091

Mathematics Complex Numbers

MCQ Focus. If $\log_2 x$ is 5, then x is:

- A. 10
- B. 25
- C. 32
- D. 64

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 32

Worked Reasoning.

$\log_2 x$ equals 5 means $x=2^5=32$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 32.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 32.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 32.

Trick / Tip. This question rewards recognising log *conversion*: rewrite the logarithm as an exponential equation; the answer usually follows in one step. The correct option should agree with 32, not just look numerically close.

Q092

Mathematics Complex Numbers

MCQ Focus. If $\log a + \log b = \log 20$ and $a=4$, then b is:

- A. 4
- B. 5
- C. 16
- D. 80

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 5

Worked Reasoning. $\log a + \log b = \log(ab)$. Thus $ab=20$, and with $a=4$, $b=5$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 5.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 5.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 5.

Trick / Tip. Question 92 is a *log conversion* item: rewrite the logarithm as an exponential equation; the answer usually follows in one step. The correct option should agree with 5, not just look numerically close.



Q093

Mathematics Complex Numbers

MCQ Focus. The solution of $2^{x+1}=16$ is:

- A. 2
- B. 3
- C. 4
- D. 5

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **B** 3

Worked Reasoning. $2^{x+1}=16=2^4$, so $x + 1 = 4$ and $x=3$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 3.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 3.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 3.

Trick / Tip. The useful shortcut here is tied to complex numbers: For complex-number MCQs, compute only the part asked for; if the question asks the real part, do not waste time finishing the imaginary part.. The correct option should agree with 3, not just look numerically close.

Q094

Mathematics Sets and Functions

MCQ Focus. If $f(x) = x^2$ and $h=0.01$, then $[f(3+h)-f(3)]/h$ equals:

- A. 6
- B. 6.01
- C. 9
- D. 0.01

Fast Route. Write the defining rule or restriction in one line; the answer usually follows without full expansion.

Trap Watch. Do not skip exclusions; domains and set differences often hide the missing restriction.

Answer. **B** 6.01

Worked Reasoning. $[f(3+h)-f(3)]/h = [(3+h)^2 - 9]/h = 6 + h = 6.01$.

Why the other options are wrong.

- ◇ Option A: misses an exclusion or restriction in the domain/set condition; it leads away from 6.01.
- ◇ Option C: counts the wrong region or keeps an element that should be removed; it leads away from 6.01.
- ◇ Option D: misses an exclusion or restriction in the domain/set condition; it leads away from 6.01.

Trick / Tip. For this Sets and Functions pattern, notice complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 6.01, not just look numerically close.



Q095

Mathematics Complex Numbers

MCQ Focus. The sum of roots of $x^3 - 4x^2 + 7x - 5 = 0$ is:

- A. -4
- B. 4
- C. 7
- D. 5

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. B 4

Worked Reasoning. For a monic cubic, sum of roots is -coefficient of x^2 . Here it is 4.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 4.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from 4.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 4.

Trick / Tip. This question rewards recognising quadratic structure: compare the requested quantity with Vieta, discriminant, or direct substitution before solving roots. The correct option should agree with 4, not just look numerically close.

Q096

Mathematics Complex Numbers

MCQ Focus. If A varies directly as B and $A=18$ when $B=6$, then A when $B=10$ is:

- A. 20
- B. 24
- C. 30
- D. 36

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 30

Worked Reasoning. Direct variation gives $A=kB$. Since $18=k(6)$, $k=3$, so $A=30$ when $B=10$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 30.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 30.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 30.

Trick / Tip. Question 96 is a complex numbers item: For complex-number MCQs, compute only the part asked for; if the question asks the real part, do not waste time finishing the imaginary part.. The correct option should agree with 30, not just look numerically close.



Q097

Mathematics Mathematics Pattern Recognition

MCQ Focus. The equation $|x+1|=4$ has solutions:

- A. 3 only
- B. -5 only
- C. 3 and -5
- D. No solution

Fast Route. Spot the tested rule first and avoid long working.

Trap Watch. The distractor usually uses the right topic but the wrong rule.

Answer. C 3 and -5

Worked Reasoning. $|x+1|=4$ gives $x+1=4$ or $x+1=-4$, hence $x=3$ or $x=-5$.

Why the other options are wrong.

- ◇ Option A: does not follow the rule used in the worked solution; it leads away from 3 and -5.
- ◇ Option B: uses a nearby but wrong pattern; it leads away from 3 and -5.
- ◇ Option D: does not follow the rule used in the worked solution; it leads away from 3 and -5.

Trick / Tip. The useful shortcut here is tied to absolute-value interval: split the inequality into a compound interval and solve both sides at once. The correct option should agree with 3 and -5, not just look numerically close.

Q098

Mathematics Quadratics and Algebra

MCQ Focus. If $x + \frac{1}{x} = 5$, then $x^2 + 1/x^2$ equals:

- A. 23
- B. 25
- C. 27
- D. 30

Fast Route. Use Vieta, discriminant, substitution, or cover-up before attempting a long solution.

Trap Watch. Sign errors and confusing sum/product of roots create most wrong options.

Answer. A 23

Worked Reasoning. Square $x + \frac{1}{x} = 5$: $x^2 + 2 + 1/x^2 = 25$, so $x^2 + 1/x^2 = 23$.

Why the other options are wrong.

- ◇ Option B: comes from a sign slip in the coefficient or discriminant step; it leads away from 23.
- ◇ Option C: stops before the algebraic simplification that the options require; it leads away from 23.
- ◇ Option D: uses the sum of roots where the product or substitution condition is being tested; it leads away from 23.

Trick / Tip. For this Quadratics and Algebra pattern, notice complex quotient: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with 23, not just look numerically close.



Q099

Mathematics Complex Numbers

MCQ Focus. The remainder when $x^3 - 2x + 5$ is divided by $x-2$ is:

- A. 1
- B. 5
- C. 9
- D. 17

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. C 9

Worked Reasoning. By the remainder theorem, substitute $x=2$: $8 - 4 + 5 = 9$.

Why the other options are wrong.

- ◇ Option A: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 9.
- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from 9.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from 9.

Trick / Tip. This question rewards recognising complex numbers: For complex-number MCQs, compute only the part asked for; if the question asks the real part, do not waste time finishing the imaginary part.. The correct option should agree with 9, not just look numerically close.

Q100

Mathematics Complex Numbers

MCQ Focus. If a line has intercepts 3 and 4 on the axes, its equation is:

- A. $\frac{x}{3} + \frac{y}{4} = 1$
- B. $\frac{x}{4} + \frac{y}{3} = 1$
- C. $3x + 4y = 1$
- D. $x + y = 7$

Fast Route. Use the cycle, conjugate, or cube-root identity first; only then compare options.

Trap Watch. The attractive wrong answer usually comes from ignoring the sign cycle or failing to rationalize.

Answer. **A** $\frac{x}{3} + \frac{y}{4} = 1$

Worked Reasoning. Intercept form is $\frac{x}{a} + \frac{y}{b} = 1$. With intercepts 3 and 4, $\frac{x}{3} + \frac{y}{4} = 1$.

Why the other options are wrong.

- ◇ Option B: selects a tempting sign or magnitude before isolating the requested real result; it leads away from $\frac{x}{3} + \frac{y}{4} = 1$.
- ◇ Option C: uses a shortcut before applying the complex-number identity completely; it leads away from $\frac{x}{3} + \frac{y}{4} = 1$.
- ◇ Option D: keeps an unreduced complex form; rationalizing or cycling powers changes this value; it leads away from $\frac{x}{3} + \frac{y}{4} = 1$.

Trick / Tip. Question 100 is a complex quotient item: multiply numerator and denominator by the conjugate; if only the real part is asked, stop once the real numerator is known. The correct option should agree with $\frac{x}{3} + \frac{y}{4} = 1$, not just look numerically close.



CHAPTER 2

Physics MCQ Strategy Lab

How to use this chapter. Use each page to identify the governing law before calculation. Unit checks and proportional reasoning often remove distractors.



Q101

Physics Measurements and Units

MCQ Focus. A measurement is recorded as 12.30 cm. Number of significant figures is:

- A. 2
- B. 3
- C. 4
- D. 5

Fast Route. Use unit/dimension rules first; they prevent unnecessary numerical work.

Trap Watch. Units that sound familiar can belong to a different physical quantity.

Answer. C 4

Worked Reasoning. 12.30 has four significant figures because trailing zeros after the decimal are significant.

Why the other options are wrong.

- ◇ Option A: counts significant figures or dimensions without applying the stated rule; it leads away from 4.
- ◇ Option B: drops a zero or power that remains significant in the measurement; it leads away from 4.
- ◇ Option D: counts significant figures or dimensions without applying the stated rule; it leads away from 4.

Trick / Tip. The useful shortcut here is tied to significant figures: count trailing zeros after a decimal as significant; do not discard measured precision. The correct option should agree with 4, not just look numerically close.



Q102

Physics Measurements and Units

MCQ Focus. The dimension of impulse is:

- A. MLT^{-1}
- B. MLT^{-2}
- C. ML^2T^{-2}
- D. $ML^{-1}T^{-2}$

Fast Route. Use unit/dimension rules first; they prevent unnecessary numerical work.

Trap Watch. Units that sound familiar can belong to a different physical quantity.

Answer. A MLT^{-1}

Worked Reasoning. Impulse equals force times time: $(MLT^{-2})T = MLT^{-1}$.

Why the other options are wrong.

- ◇ Option B: drops a zero or power that remains significant in the measurement; it leads away from MLT^{-1} .
- ◇ Option C: matches a familiar-looking unit idea but not the tested quantity; it leads away from MLT^{-1} .
- ◇ Option D: counts significant figures or dimensions without applying the stated rule; it leads away from MLT^{-1} .

Trick / Tip. For this Measurements and Units pattern, notice dimensional formula: derive the dimension from the physical definition instead of memorising isolated units. The correct option should agree with MLT^{-1} , not just look numerically close.

Q103

Physics Measurements and Units

MCQ Focus. If percentage errors in length and width are 2% and 3%, percentage error in area is approximately:

- A. 1%
- B. 5%
- C. 6%
- D. 9%

Fast Route. Use unit/dimension rules first; they prevent unnecessary numerical work.

Trap Watch. Units that sound familiar can belong to a different physical quantity.

Answer. **B** 5%

Worked Reasoning. For $\text{area} = \text{length} \times \text{width}$, percentage errors add: 2 percent + 3 percent = 5 percent.

Why the other options are wrong.

- ◇ Option A: counts significant figures or dimensions without applying the stated rule; it leads away from 5%.
- ◇ Option C: matches a familiar-looking unit idea but not the tested quantity; it leads away from 5%.
- ◇ Option D: counts significant figures or dimensions without applying the stated rule; it leads away from 5%.

Trick / Tip. This question rewards recognising arithmetic progression: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with 5%, not just look numerically close.



Q104

Physics Mechanics

MCQ Focus. Two vectors of magnitudes 5 and 12 are perpendicular. Resultant magnitude is:

- A. 7
- B. 13
- C. 17
- D. 60

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. B 13

Worked Reasoning. For perpendicular vectors, $\text{resultant} = \sqrt{5^2 + 12^2} = 13$.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from 13.
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from 13.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from 13.

Trick / Tip. Question 104 is a perpendicular-vector resultant item: use the Pythagorean result because perpendicular vectors form a right triangle. The correct option should agree with 13, not just look numerically close.

Q105

Physics Mechanics

MCQ Focus. A 10 N force makes angle 60° with displacement 4 m. Work done is:

- A. 10 J
- B. 20 J
- C. 40 J
- D. 80 J

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. **B** 20 J

Worked Reasoning. Work $W = Fs \cos \theta = 10 \times 4 \times \cos 60^\circ = 20$ J.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from 20 J.
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from 20 J.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from 20 J.

Trick / Tip. The useful shortcut here is tied to work component: use $W = Fs \cos \theta$; perpendicular components do not contribute to work. The correct option should agree with 20 J, not just look numerically close.



Q106

Physics Mechanics

MCQ Focus. If a body moves with constant speed in a circle, its acceleration is:

- A. Zero
- B. Tangential
- C. Centripetal
- D. Along velocity

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. ☒ Centripetal

Worked Reasoning. Uniform circular motion has acceleration toward the center, called centripetal acceleration.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Centripetal.
- ◇ Option B: misses the square/proportionality relationship in the law; it leads away from Centripetal.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Centripetal.

Trick / Tip. For this Mechanics pattern, notice trigonometric ratio: use the triangle ratio implied by the given function, then derive the requested function. The correct option should agree with Centripetal, not just look numerically close.



Q107

Physics Electricity and Magnetism

MCQ Focus. A car starts from rest and accelerates at 2 m/s^2 for 5 s. Distance covered is:

- A. 10 m
- B. 20 m
- C. 25 m
- D. 50 m

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. C 25 m

Worked Reasoning. From rest, $s = ut + \left(\frac{1}{2}\right)at^2 = 0 + \left(\frac{1}{2}\right)(2)(5^2) = 25 \text{ m}$.

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from 25 m.
- ◇ Option B: uses the wrong circuit, field, or AC formula; it leads away from 25 m.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from 25 m.

Trick / Tip. This question rewards recognising constant-acceleration distance: from rest, use $s = ut + \text{one-half at squared}$ with $u=0$. The correct option should agree with 25 m, not just look numerically close.



Q108

Physics Mechanics

MCQ Focus. A projectile has maximum range when angle of projection is:

- A. 30°
- B. 45°
- C. 60°
- D. 90°

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. B 45°

Worked Reasoning. Projectile range is maximum at 45° when launch and landing heights are the same.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from 45° .
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from 45° .
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from 45° .

Trick / Tip. Question 108 is a mechanics item: For work, check the component along displacement: $W = F_s \cos \theta$, not simply force times distance.. The correct option should agree with 45° , not just look numerically close.

Q109

Physics Mechanics

MCQ Focus. If momentum doubles while mass remains constant, kinetic energy becomes:

- A. Half
- B. Double
- C. Four times
- D. Unchanged

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. ☒ C Four times

Worked Reasoning. $K = p^2 / (2m)$. If p doubles and m is constant, kinetic energy becomes four times.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Four times.
- ◇ Option B: misses the square/proportionality relationship in the law; it leads away from Four times.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Four times.

Trick / Tip. The useful shortcut here is tied to momentum-energy proportionality: at constant mass, kinetic energy is proportional to momentum squared. The correct option should agree with Four times, not just look numerically close.



Q110

Physics Waves and Fluids

MCQ Focus. Escape velocity from a planet depends on:

- A. Mass and radius of planet
- B. Mass of escaping body
- C. Temperature only
- D. Air pressure only

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. **A** Mass and radius of planet

Worked Reasoning. Escape velocity $v_e = \sqrt{2GM/R}$, so it depends on the planet mass and radius.

Why the other options are wrong.

- ◊ Option B: skips the defining relation that links the quantities; it leads away from Mass and radius of planet.
- ◊ Option C: uses a qualitative statement that does not match the physical condition; it leads away from Mass and radius of planet.
- ◊ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Mass and radius of planet.

Trick / Tip. For this Waves and Fluids pattern, notice arithmetic progression: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with Mass and radius of planet, not just look numerically close.

Q111

Physics Mechanics

MCQ Focus. For SHM, acceleration is maximum at:

- A. Mean position
- B. Extreme position
- C. Half amplitude
- D. All positions

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. **B** Extreme position

Worked Reasoning. In SHM, acceleration magnitude is $\omega^2 x$, so it is maximum at extreme displacement.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Extreme position.
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from Extreme position.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Extreme position.

Trick / Tip. This question rewards recognising mechanics: For work, check the component along displacement: $W = Fs \cos \theta$, not simply force times distance.. The correct option should agree with Extreme position, not just look numerically close.

Q112

Physics Physics Law Recognition

MCQ Focus. The time period of a simple pendulum is independent of:

- A. Length
- B. Gravity
- C. Small amplitude mass
- D. Local value of g

Fast Route. Spot the tested rule first and avoid long working.

Trap Watch. The distractor usually uses the right topic but the wrong rule.

Answer. C Small amplitude mass

Worked Reasoning. For a simple pendulum, $T=2\pi \sqrt{l/g}$, independent of bob mass for small oscillations.

Why the other options are wrong.

- ◇ Option A: does not follow the rule used in the worked solution; it leads away from Small amplitude mass.
- ◇ Option B: uses a nearby but wrong pattern; it leads away from Small amplitude mass.
- ◇ Option D: does not follow the rule used in the worked solution; it leads away from Small amplitude mass.

Trick / Tip. Question 112 is a physics law recognition item: Use the tested rule for physics law recognition, then compare against the options.. The correct option should agree with Small amplitude mass, not just look numerically close.

Q113

Physics Waves and Fluids

MCQ Focus. If frequency of a wave is 50 Hz and wavelength is 4 m, speed is:

- A. 12.5 m/s
- B. 54 m/s
- C. 200 m/s
- D. 400 m/s

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. C 200 m/s

Worked Reasoning. Wave speed $v = f \lambda = 50 \cdot 4 = 200$ m/s.

Why the other options are wrong.

- ◇ Option A: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from 200 m/s.
- ◇ Option B: skips the defining relation that links the quantities; it leads away from 200 m/s.
- ◇ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from 200 m/s.

Trick / Tip. The useful shortcut here is tied to wave relation: apply $v = f \lambda$ and check that the units become metres per second. The correct option should agree with 200 m/s, not just look numerically close.



Q114

Physics Waves and Fluids

MCQ Focus. Sound waves in air are mainly:

- A. Transverse
- B. Longitudinal
- C. Electromagnetic
- D. Stationary only

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. **B** Longitudinal

Worked Reasoning. Sound in air travels by compressions and rarefactions, so it is longitudinal.

Why the other options are wrong.

- ◇ Option A: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Longitudinal.
- ◇ Option C: uses a qualitative statement that does not match the physical condition; it leads away from Longitudinal.
- ◇ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Longitudinal.

Trick / Tip. For this Waves and Fluids pattern, notice wave relation: apply $v=f\lambda$ and check that the units become metres per second. The correct option should agree with Longitudinal, not just look numerically close.



Q115

Physics Waves and Fluids

MCQ Focus. Doppler effect is observed when there is relative motion between:

- A. Source and observer
- B. Two charges only
- C. Two lenses only
- D. Two mirrors only

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. **A** Source and observer

Worked Reasoning. The Doppler effect occurs due to relative motion between source and observer.

Why the other options are wrong.

- ◇ Option B: skips the defining relation that links the quantities; it leads away from Source and observer.
- ◇ Option C: uses a qualitative statement that does not match the physical condition; it leads away from Source and observer.
- ◇ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Source and observer.

Trick / Tip. This question rewards recognising waves and fluids: For sound and Doppler items, identify relative motion before looking at formulas.. The correct option should agree with Source and observer, not just look numerically close.



Q116

Physics Waves and Fluids

MCQ Focus. The SI unit of pressure is:

- A. Watt
- B. Pascal
- C. Joule
- D. Tesla

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. **B** Pascal

Worked Reasoning. The SI unit of pressure is the pascal.

Why the other options are wrong.

- ◇ Option A: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Pascal.
- ◇ Option C: uses a qualitative statement that does not match the physical condition; it leads away from Pascal.
- ◇ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Pascal.

Trick / Tip. Question 116 is a pressure/fluid concept item: identify the fluid law or pressure definition before doing any calculation. The correct option should agree with Pascal, not just look numerically close.

Q117

Physics Waves and Fluids

MCQ Focus. Terminal velocity of a small sphere in viscous liquid increases when radius:

- A. Decreases
- B. Increases
- C. Becomes zero
- D. Has no effect

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. **B** Increases

Worked Reasoning. Terminal velocity in Stokes law is proportional to r^2 , so it increases with radius.

Why the other options are wrong.

- ◇ Option A: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Increases.
- ◇ Option C: uses a qualitative statement that does not match the physical condition; it leads away from Increases.
- ◇ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Increases.

Trick / Tip. The useful shortcut here is tied to waves and fluids: Use $v=f\lambda$ as a unit check: Hz times metres must give metres per second.. The correct option should agree with Increases, not just look numerically close.



Q118

Physics Mechanics

MCQ Focus. Bernoulli's equation is based on conservation of:

- A. Mass only
- B. Energy
- C. Charge
- D. Momentum only

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. **B** Energy

Worked Reasoning. Bernoulli equation expresses conservation of mechanical energy in fluid flow.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Energy.
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from Energy.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Energy.

Trick / Tip. For this Mechanics pattern, notice momentum relation: For kinetic energy and momentum, remember $KE = p^2/(2m)$; doubling momentum quadruples kinetic energy at constant mass.. The correct option should agree with Energy, not just look numerically close.

Q119

Physics Mechanics

MCQ Focus. In an isothermal process for an ideal gas, internal energy change is:

- A. Positive
- B. Negative
- C. Zero
- D. Infinite

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. ☒ C Zero

Worked Reasoning. For an ideal gas, internal energy depends only on temperature; isothermal means no temperature change.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Zero.
- ◇ Option B: misses the square/proportionality relationship in the law; it leads away from Zero.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Zero.

Trick / Tip. This question rewards recognising energy relation: choose the conservation or energy formula named by the stem, not a nearby force relation. The correct option should agree with Zero, not just look numerically close.



Q120

Physics Mechanics

MCQ Focus. Efficiency of a Carnot engine depends on:

- A. Working substance
- B. Reservoir temperatures
- C. Piston area
- D. Gas volume only

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. **B** Reservoir temperatures

Worked Reasoning. Carnot efficiency depends only on hot and cold reservoir temperatures.

Why the other options are wrong.

- ◊ Option A: uses a formula for a different mechanical quantity; it leads away from Reservoir temperatures.
- ◊ Option C: uses the right data but the wrong component or direction; it leads away from Reservoir temperatures.
- ◊ Option D: uses a formula for a different mechanical quantity; it leads away from Reservoir temperatures.

Trick / Tip. Question 120 is a trigonometric ratio item: use the triangle ratio implied by the given function, then derive the requested function. The correct option should agree with Reservoir temperatures, not just look numerically close.

Q121

Physics Thermal Physics

MCQ Focus. Heat required to change phase at constant temperature is called:

- A. Specific heat
- B. Latent heat
- C. Thermal capacity
- D. Conductivity

Fast Route. Identify the process or law before using numbers; thermal MCQs often test concepts.

Trap Watch. A process condition such as isothermal or constant temperature changes the result.

Answer. **B** Latent heat

Worked Reasoning. Heat used for phase change at constant temperature is latent heat.

Why the other options are wrong.

- ◇ Option A: ignores the thermal process named in the stem; it leads away from Latent heat.
- ◇ Option C: confuses heat, internal energy, temperature, and efficiency; it leads away from Latent heat.
- ◇ Option D: ignores the thermal process named in the stem; it leads away from Latent heat.

Trick / Tip. The useful shortcut here is tied to arithmetic progression: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with Latent heat, not just look numerically close.



Q122

Physics Mechanics

MCQ Focus. The first law of thermodynamics is a statement of conservation of:

- A. Charge
- B. Mass
- C. Energy
- D. Momentum only

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. ☒ C Energy

Worked Reasoning. The first law of thermodynamics is conservation of energy applied to heat and work.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Energy.
- ◇ Option B: misses the square/proportionality relationship in the law; it leads away from Energy.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Energy.

Trick / Tip. For this Mechanics pattern, notice momentum relation: For projectile range, the maximum range angle in ideal motion is 45° . The correct option should agree with Energy, not just look numerically close.

Q123

Physics Electricity and Magnetism

MCQ Focus. Electric field due to a point charge varies with distance r as:

- A. r
- B. $1/r$
- C. $1/r^2$
- D. r^2

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. ☒ C $1/r^2$

Worked Reasoning. Electric field of a point charge is $E=kq/r^2$, so it varies as $1/r^2$.

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from $1/r^2$.
- ◇ Option B: uses the wrong circuit, field, or AC formula; it leads away from $1/r^2$.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from $1/r^2$.

Trick / Tip. This question rewards recognising trigonometric ratio: use the triangle ratio implied by the given function, then derive the requested function. The correct option should agree with $1/r^2$, not just look numerically close.

Q124

Physics Electricity and Magnetism

MCQ Focus. The SI unit of electric potential is:

- A. Coulomb
- B. Volt
- C. Ohm
- D. Farad

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. **B** Volt

Worked Reasoning. The SI unit of electric potential is the volt.

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from Volt.
- ◇ Option C: forgets the condition for zero, maximum, resonance, or induction; it leads away from Volt.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from Volt.

Trick / Tip. Question 124 is a Ohm law item: use $I=V/R$ and keep volts, ohms, and amperes in their roles. The correct option should agree with Volt, not just look numerically close.



Q125

Physics Electricity and Magnetism

MCQ Focus. Two equal capacitors C in series have equivalent capacitance:

- A. $2C$
- B. C
- C. $C/2$
- D. C^2

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. ☒ C $C/2$

Worked Reasoning. For two equal capacitors in series, $1/C_{eq} = \frac{1}{C} + \frac{1}{C} = 2/C$, so $C_{eq} = C/2$.

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from $C/2$.
- ◇ Option B: uses the wrong circuit, field, or AC formula; it leads away from $C/2$.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from $C/2$.

Trick / Tip. The useful shortcut here is tied to arithmetic progression: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with $C/2$, not just look numerically close.



Q126

Physics Electricity and Magnetism

MCQ Focus. If distance between parallel capacitor plates is doubled, capacitance becomes:

- A. Double
- B. Half
- C. Four times
- D. Unchanged

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. **B** Half

Worked Reasoning. Parallel-plate capacitance is $C = \epsilon A/d$. Doubling d halves C .

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from Half.
- ◇ Option C: forgets the condition for zero, maximum, resonance, or induction; it leads away from Half.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from Half.

Trick / Tip. For this Electricity and Magnetism pattern, notice arithmetic progression: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with Half, not just look numerically close.



Q127

Physics Thermal Physics

MCQ Focus. Ohm's law is valid when:

- A. Temperature and physical conditions remain constant
- B. Current is zero
- C. Voltage is zero
- D. Resistance changes rapidly

Fast Route. Identify the process or law before using numbers; thermal MCQs often test concepts.

Trap Watch. A process condition such as isothermal or constant temperature changes the result.

Answer. **A** Temperature and physical conditions remain constant

Worked Reasoning. Ohm law $V=IR$ applies when temperature and other physical conditions stay constant.

Why the other options are wrong.

- ◇ Option B: applies a heat-engine or gas-law idea to the wrong condition; it leads away from Temperature and physical conditions remain constant.
- ◇ Option C: confuses heat, internal energy, temperature, and efficiency; it leads away from Temperature and physical conditions remain constant.
- ◇ Option D: ignores the thermal process named in the stem; it leads away from Temperature and physical conditions remain constant.

Trick / Tip. This question rewards recognising arithmetic progression: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with Temperature and physical conditions remain constant, not just look numerically close.



Q128

Physics Electricity and Magnetism

MCQ Focus. A 12 V battery connected to 3 ohm resistor gives current:

- A. 2 A
- B. 3 A
- C. 4 A
- D. 36 A

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. C 4 A

Worked Reasoning. Use $I = \frac{V}{R} = \frac{12}{3} = 4$ A.

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from 4 A.
- ◇ Option B: uses the wrong circuit, field, or AC formula; it leads away from 4 A.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from 4 A.

Trick / Tip. Question 128 is a resistance law item: use Ohm law or equivalent resistance after identifying series or parallel arrangement. The correct option should agree with 4 A, not just look numerically close.

Q129

Physics Electricity and Magnetism

MCQ Focus. Three 6 ohm resistors in parallel have equivalent resistance:

- A. 2 ohm
- B. 3 ohm
- C. 6 ohm
- D. 18 ohm

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. A 2 ohm

Worked Reasoning. Three equal 6 ohm resistors in parallel give $R_{eq} = \frac{R}{3} = 2$ ohm.

Why the other options are wrong.

- ◇ Option B: uses the wrong circuit, field, or AC formula; it leads away from 2 ohm.
- ◇ Option C: forgets the condition for zero, maximum, resonance, or induction; it leads away from 2 ohm.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from 2 ohm.

Trick / Tip. The useful shortcut here is tied to trigonometric ratio: use the triangle ratio implied by the given function, then derive the requested function. The correct option should agree with 2 ohm, not just look numerically close.



Q130

Physics Electricity and Magnetism

MCQ Focus. Power dissipated in a 4 ohm resistor carrying 3 A is:

- A. 12 W
- B. 24 W
- C. 36 W
- D. 48 W

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. C 36 W

Worked Reasoning. Power $P = I^2 R = 3^2 \cdot 4 = 36$ W.

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from 36 W.
- ◇ Option B: uses the wrong circuit, field, or AC formula; it leads away from 36 W.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from 36 W.

Trick / Tip. For this Electricity and Magnetism pattern, notice resistance law: use Ohm law or equivalent resistance after identifying series or parallel arrangement. The correct option should agree with 36 W, not just look numerically close.

Q131

Physics Mechanics

MCQ Focus. Magnetic force on a charge moving parallel to magnetic field is:

- A. Maximum
- B. Zero
- C. qvB
- D. qB/v

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. **B** Zero

Worked Reasoning. Magnetic force $qvB \sin \theta$ is zero when velocity is parallel to the magnetic field.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Zero.
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from Zero.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Zero.

Trick / Tip. This question rewards recognising force/work component: use $W = Fs \cos \theta$ and include only the component of force along displacement. The correct option should agree with Zero, not just look numerically close.

Q132

Physics Electricity and Magnetism

MCQ Focus. The unit of magnetic flux is:

- A. Tesla
- B. Weber
- C. Henry
- D. Farad

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. **B** Weber

Worked Reasoning. The SI unit of magnetic flux is the weber.

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from Weber.
- ◇ Option C: forgets the condition for zero, maximum, resonance, or induction; it leads away from Weber.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from Weber.

Trick / Tip. Question 132 is a magnetic-force condition item: magnetic force depends on $\sin \theta$; parallel motion gives zero force. The correct option should agree with Weber, not just look numerically close.



Q133

Physics Mechanics

MCQ Focus. A current-carrying wire in magnetic field experiences maximum force when angle between current and field is:

- A. 0°
- B. 30°
- C. 60°
- D. 90°

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. D 90°

Worked Reasoning. Magnetic force on a current is $BIL \sin \theta$, maximum at 90° .

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from 90° .
- ◇ Option B: misses the square/proportionality relationship in the law; it leads away from 90° .
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from 90° .

Trick / Tip. The useful shortcut here is tied to force/work component: use $W = F_s \cos \theta$ and include only the component of force along displacement. The correct option should agree with 90° , not just look numerically close.



Q134

Physics Electricity and Magnetism

MCQ Focus. The device used to convert AC to DC is:

- A. Transformer
- B. Rectifier
- C. Inductor
- D. Capacitor

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. **B** Rectifier

Worked Reasoning. A rectifier converts AC into DC.

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from Rectifier.
- ◇ Option C: forgets the condition for zero, maximum, resonance, or induction; it leads away from Rectifier.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from Rectifier.

Trick / Tip. For this Electricity and Magnetism pattern, notice arithmetic progression: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with Rectifier, not just look numerically close.

Q135

Physics Mechanics

MCQ Focus. A transformer works on:

- A. Coulomb induction
- B. Mutual induction
- C. Photoelectric effect
- D. Thermionic emission

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. **B** Mutual induction

Worked Reasoning. A transformer operates by mutual induction between coils.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Mutual induction.
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from Mutual induction.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Mutual induction.

Trick / Tip. This question rewards recognising induction principle: a transformer needs changing magnetic flux, so DC fails after the switching instant. The correct option should agree with Mutual induction, not just look numerically close.

Q136

Physics Waves and Fluids

MCQ Focus. A transformer cannot operate on DC because:

- A. DC has no voltage
- B. DC has no changing magnetic flux after switching
- C. DC has infinite frequency
- D. Core becomes weightless

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. **B** DC has no changing magnetic flux after switching

Worked Reasoning. Steady DC gives no continuously changing magnetic flux, so a transformer cannot operate normally on DC.

Why the other options are wrong.

- ◊ Option A: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from DC has no changing magnetic flux after switching.
- ◊ Option C: uses a qualitative statement that does not match the physical condition; it leads away from DC has no changing magnetic flux after switching.
- ◊ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from DC has no changing magnetic flux after switching.

Trick / Tip. Question 136 is a frequency relation item: connect frequency and wavelength through speed, not by adding the quantities. The correct option should agree with DC has no changing magnetic flux after switching, not just look numerically close.



Q137

Physics Electricity and Magnetism

MCQ Focus. The reactance of a capacitor in AC circuit is:

- A. $2\pi fC$
- B. $1/(2\pi fC)$
- C. f/C
- D. C/f

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. B $1/(2\pi fC)$

Worked Reasoning. Capacitive reactance is $X_C = 1/(2\pi fC)$.

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from $1/(2\pi fC)$.
- ◇ Option C: forgets the condition for zero, maximum, resonance, or induction; it leads away from $1/(2\pi fC)$.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from $1/(2\pi fC)$.

Trick / Tip. The useful shortcut here is tied to arithmetic progression: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with $1/(2\pi fC)$, not just look numerically close.



Q138

Physics Electricity and Magnetism

MCQ Focus. In pure inductor AC circuit, current:

- A. Leads voltage by 90°
- B. Lags voltage by 90°
- C. Is in phase
- D. Is zero always

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. **B** Lags voltage by 90°

Worked Reasoning. In a pure inductor, current lags voltage by 90° .

Why the other options are wrong.

- ◊ Option A: reverses a series/parallel or phase relationship; it leads away from Lags voltage by 90° .
- ◊ Option C: forgets the condition for zero, maximum, resonance, or induction; it leads away from Lags voltage by 90° .
- ◊ Option D: reverses a series/parallel or phase relationship; it leads away from Lags voltage by 90° .

Trick / Tip. For this Electricity and Magnetism pattern, notice electricity and magnetism: For parallel resistors or capacitors, ask whether the equivalent moves smaller or larger than each component.. The correct option should agree with Lags voltage by 90° , not just look numerically close.



Q139

Physics Electricity and Magnetism

MCQ Focus. Resonance in LCR circuit occurs when:

- A. $X_L = X_C$
- B. $R=0$ only
- C. $f=0$
- D. $V=0$

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. A $X_L = X_C$

Worked Reasoning. LCR resonance occurs when inductive reactance equals capacitive reactance.

Why the other options are wrong.

- ◇ Option B: uses the wrong circuit, field, or AC formula; it leads away from $X_L = X_C$.
- ◇ Option C: forgets the condition for zero, maximum, resonance, or induction; it leads away from $X_L = X_C$.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from $X_L = X_C$.

Trick / Tip. This question rewards recognising electricity and magnetism: For AC phase questions, memorise the direction: inductor lags current, capacitor leads current depending on wording.. The correct option should agree with $X_L = X_C$, not just look numerically close.

Q140

Physics Mechanics

MCQ Focus. Young's modulus is the ratio of:

- A. Stress to strain
- B. Strain to stress
- C. Force to velocity
- D. Power to area

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. A Stress to strain

Worked Reasoning. Young modulus is defined as stress divided by strain.

Why the other options are wrong.

- ◇ Option B: misses the square/proportionality relationship in the law; it leads away from Stress to strain.
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from Stress to strain.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Stress to strain.

Trick / Tip. Question 140 is a force/work component item: use $W = F_s \cos \theta$ and include only the component of force along displacement. The correct option should agree with Stress to strain, not just look numerically close.

Q141

Physics Modern Physics

MCQ Focus. Semiconductors have conductivity:

- A. Less than insulators
- B. Between conductors and insulators
- C. Infinite
- D. Always zero

Fast Route. Recall the defining formula or fact; modern physics items are usually recognition-led.

Trap Watch. Do not replace the defining concept with a classical analogy.

Answer. **B** Between conductors and insulators

Worked Reasoning. Semiconductors conduct better than insulators but worse than conductors.

Why the other options are wrong.

- ◇ Option A: replaces the defining modern-physics fact with a classical-looking idea; it leads away from Between conductors and insulators.
- ◇ Option C: matches a familiar term but not the phenomenon in the stem; it leads away from Between conductors and insulators.
- ◇ Option D: replaces the defining modern-physics fact with a classical-looking idea; it leads away from Between conductors and insulators.

Trick / Tip. The useful shortcut here is tied to carrier concept: for semiconductors, identify majority carriers before thinking about current direction. The correct option should agree with Between conductors and insulators, not just look numerically close.



Q142

Physics Modern Physics

MCQ Focus. In a p-type semiconductor, majority carriers are:

- A. Electrons
- B. Holes
- C. Protons
- D. Photons

Fast Route. Recall the defining formula or fact; modern physics items are usually recognition-led.

Trap Watch. Do not replace the defining concept with a classical analogy.

Answer. B Holes

Worked Reasoning. In p-type material, holes are the majority carriers.

Why the other options are wrong.

- ◇ Option A: replaces the defining modern-physics fact with a classical-looking idea; it leads away from Holes.
- ◇ Option C: matches a familiar term but not the phenomenon in the stem; it leads away from Holes.
- ◇ Option D: replaces the defining modern-physics fact with a classical-looking idea; it leads away from Holes.

Trick / Tip. For this Modern Physics pattern, notice carrier concept: for semiconductors, identify majority carriers before thinking about current direction. The correct option should agree with Holes, not just look numerically close.

Q143

Physics Modern Physics

MCQ Focus. A forward-biased diode has:

- A. High resistance
- B. Low resistance
- C. Infinite resistance
- D. Negative mass

Fast Route. Recall the defining formula or fact; modern physics items are usually recognition-led.

Trap Watch. Do not replace the defining concept with a classical analogy.

Answer. **B** Low resistance

Worked Reasoning. Forward bias reduces the diode barrier and gives low resistance.

Why the other options are wrong.

- ◇ Option A: replaces the defining modern-physics fact with a classical-looking idea; it leads away from Low resistance.
- ◇ Option C: matches a familiar term but not the phenomenon in the stem; it leads away from Low resistance.
- ◇ Option D: replaces the defining modern-physics fact with a classical-looking idea; it leads away from Low resistance.

Trick / Tip. This question rewards recognising trigonometric ratio: use the triangle ratio implied by the given function, then derive the requested function. The correct option should agree with Low resistance, not just look numerically close.



Q144

Physics Electricity and Magnetism

MCQ Focus. A transistor can be used as:

- A. Amplifier
- B. Rectifier only
- C. Transformer only
- D. Ammeter only

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. A Amplifier

Worked Reasoning. A transistor is commonly used as an amplifier.

Why the other options are wrong.

- ◊ Option B: uses the wrong circuit, field, or AC formula; it leads away from Amplifier.
- ◊ Option C: forgets the condition for zero, maximum, resonance, or induction; it leads away from Amplifier.
- ◊ Option D: reverses a series/parallel or phase relationship; it leads away from Amplifier.

Trick / Tip. Question 144 is a induction principle item: a transformer needs changing magnetic flux, so DC fails after the switching instant. The correct option should agree with Amplifier, not just look numerically close.

Q145

Physics Waves and Fluids

MCQ Focus. Photoelectric emission supports the particle nature of:

- A. Sound
- B. Light
- C. Heat only
- D. Magnetism only

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. **B** Light

Worked Reasoning. Photoelectric emission supports the particle nature of light through photon energy.

Why the other options are wrong.

- ◇ Option A: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Light.
- ◇ Option C: uses a qualitative statement that does not match the physical condition; it leads away from Light.
- ◇ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Light.

Trick / Tip. The useful shortcut here is tied to photon-energy idea: photoelectric stopping potential follows photon frequency, not intensity alone. The correct option should agree with Light, not just look numerically close.



Q146

Physics Waves and Fluids

MCQ Focus. Stopping potential in photoelectric effect depends on:

- A. Intensity only
- B. Frequency of incident light
- C. Area only
- D. Pressure only

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. **B** Frequency of incident light

Worked Reasoning. Stopping potential depends on the maximum kinetic energy, which depends on frequency.

Why the other options are wrong.

- ◊ Option A: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Frequency of incident light.
- ◊ Option C: uses a qualitative statement that does not match the physical condition; it leads away from Frequency of incident light.
- ◊ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Frequency of incident light.

Trick / Tip. For this Waves and Fluids pattern, notice frequency relation: connect frequency and wavelength through speed, not by adding the quantities. The correct option should agree with Frequency of incident light, not just look numerically close.



Q147

Physics Waves and Fluids

MCQ Focus. de Broglie wavelength of a particle is:

- A. h/p
- B. hp
- C. p/h
- D. hE

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. A h/p

Worked Reasoning. de Broglie wavelength is $\lambda = h/p$.

Why the other options are wrong.

- ◇ Option B: skips the defining relation that links the quantities; it leads away from h/p .
- ◇ Option C: uses a qualitative statement that does not match the physical condition; it leads away from h/p .
- ◇ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from h/p .

Trick / Tip. This question rewards recognising wave relation: apply $v = f\lambda$ and check that the units become metres per second. The correct option should agree with h/p , not just look numerically close.



Q148

Physics Mechanics

MCQ Focus. The rest mass energy of a particle is:

- A. mc
- B. mc^2
- C. m/c
- D. c^2/m

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. B mc^2

Worked Reasoning. Rest mass energy is given by $E=mc^2$.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from mc^2 .
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from mc^2 .
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from mc^2 .

Trick / Tip. Question 148 is a energy relation item: choose the conservation or energy formula named by the stem, not a nearby force relation. The correct option should agree with mc^2 , not just look numerically close.

Q149

Physics Mechanics

MCQ Focus. Bohr's atomic model quantizes:

- A. Only mass
- B. Angular momentum
- C. Temperature
- D. Pressure

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. **B** Angular momentum

Worked Reasoning. Bohr model quantizes the angular momentum of the electron.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Angular momentum.
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from Angular momentum.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Angular momentum.

Trick / Tip. The useful shortcut here is tied to momentum relation: For projectile range, the maximum range angle in ideal motion is 45° . The correct option should agree with Angular momentum, not just look numerically close.



Q150

Physics Mechanics

MCQ Focus. Ionization energy of hydrogen atom is approximately:

- A. 3.4 eV
- B. 10.2 eV
- C. 13.6 eV
- D. 27.2 eV

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. C 13.6 eV

Worked Reasoning. The ionization energy of hydrogen in the ground state is about 13.6 eV.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from 13.6 eV.
- ◇ Option B: misses the square/proportionality relationship in the law; it leads away from 13.6 eV.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from 13.6 eV.

Trick / Tip. For this Mechanics pattern, notice arithmetic progression: use $a_n = a + (n-1)d$ and avoid summing terms unless the question asks for a sum. The correct option should agree with 13.6 eV, not just look numerically close.

Q151

Physics Waves and Fluids

MCQ Focus. Balmer series of hydrogen lies mainly in:

- A. Visible region
- B. Infrared region
- C. Microwave region
- D. Radio region

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. **A** Visible region

Worked Reasoning. The Balmer series lies mainly in the visible region.

Why the other options are wrong.

- ◇ Option B: skips the defining relation that links the quantities; it leads away from Visible region.
- ◇ Option C: uses a qualitative statement that does not match the physical condition; it leads away from Visible region.
- ◇ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Visible region.

Trick / Tip. This question rewards recognising wave relation: apply $v=f\lambda$ and check that the units become metres per second. The correct option should agree with Visible region, not just look numerically close.



Q152

Physics Modern Physics

MCQ Focus. The nucleus contains:

- A. Electrons only
- B. Protons and neutrons
- C. Photons only
- D. Neutrons only

Fast Route. Recall the defining formula or fact; modern physics items are usually recognition-led.

Trap Watch. Do not replace the defining concept with a classical analogy.

Answer. **B** Protons and neutrons

Worked Reasoning. The nucleus contains protons and neutrons.

Why the other options are wrong.

- ◇ Option A: replaces the defining modern-physics fact with a classical-looking idea; it leads away from Protons and neutrons.
- ◇ Option C: matches a familiar term but not the phenomenon in the stem; it leads away from Protons and neutrons.
- ◇ Option D: replaces the defining modern-physics fact with a classical-looking idea; it leads away from Protons and neutrons.

Trick / Tip. Question 152 is a nuclear composition item: separate α , β , γ , protons, and neutrons by composition before choosing. The correct option should agree with Protons and neutrons, not just look numerically close.

Q153

Physics Modern Physics

MCQ Focus. α particle is:

- A. Electron
- B. Helium nucleus
- C. Photon
- D. Neutron

Fast Route. Recall the defining formula or fact; modern physics items are usually recognition-led.

Trap Watch. Do not replace the defining concept with a classical analogy.

Answer. **B** Helium nucleus

Worked Reasoning. An α particle is a helium nucleus.

Why the other options are wrong.

- ◇ Option A: replaces the defining modern-physics fact with a classical-looking idea; it leads away from Helium nucleus.
- ◇ Option C: matches a familiar term but not the phenomenon in the stem; it leads away from Helium nucleus.
- ◇ Option D: replaces the defining modern-physics fact with a classical-looking idea; it leads away from Helium nucleus.

Trick / Tip. The useful shortcut here is tied to nuclear composition: separate α , β , γ , protons, and neutrons by composition before choosing. The correct option should agree with Helium nucleus, not just look numerically close.



Q154

Physics Modern Physics

MCQ Focus. The most penetrating common nuclear radiation is:

- A. α
- B. β
- C. γ
- D. Proton beam always

Fast Route. Recall the defining formula or fact; modern physics items are usually recognition-led.

Trap Watch. Do not replace the defining concept with a classical analogy.

Answer. C γ

Worked Reasoning. γ radiation is the most penetrating among common α , β , and γ radiations.

Why the other options are wrong.

- ◊ Option A: replaces the defining modern-physics fact with a classical-looking idea; it leads away from γ .
- ◊ Option B: uses the wrong particle, radiation, or atomic model relation; it leads away from γ .
- ◊ Option D: replaces the defining modern-physics fact with a classical-looking idea; it leads away from γ .

Trick / Tip. For this Modern Physics pattern, notice modern physics: recall the exact defining relation before using a classical analogy. The correct option should agree with γ , not just look numerically close.

Q155

Physics Electricity and Magnetism

MCQ Focus. Half-life is the time in which:

- A. All nuclei decay
- B. Half the undecayed nuclei decay
- C. Mass becomes zero
- D. Activity doubles

Fast Route. Choose the circuit or field law first, then substitute; series/parallel and phase conditions are common shortcuts.

Trap Watch. Series and parallel formulas, and AC phase statements, are common reversals.

Answer. **B** Half the undecayed nuclei decay

Worked Reasoning. Half-life is the time required for half the undecayed nuclei to decay.

Why the other options are wrong.

- ◇ Option A: reverses a series/parallel or phase relationship; it leads away from Half the undecayed nuclei decay.
- ◇ Option C: forgets the condition for zero, maximum, resonance, or induction; it leads away from Half the undecayed nuclei decay.
- ◇ Option D: reverses a series/parallel or phase relationship; it leads away from Half the undecayed nuclei decay.

Trick / Tip. This question rewards recognising electricity and magnetism: For magnetic force, parallel motion gives zero force because $\sin \theta$ is zero.. The correct option should agree with Half the undecayed nuclei decay, not just look numerically close.



Q156

Physics Mechanics

MCQ Focus. Binding energy per nucleon is maximum near:

- A. Hydrogen
- B. Iron
- C. Uranium
- D. Helium only

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. **B** Iron

Worked Reasoning. Binding energy per nucleon is maximum near iron.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Iron.
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from Iron.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Iron.

Trick / Tip. Question 156 is a energy relation item: choose the conservation or energy formula named by the stem, not a nearby force relation. The correct option should agree with Iron, not just look numerically close.

Q157

Physics Waves and Fluids

MCQ Focus. Nuclear fission is generally splitting of:

- A. Light nucleus into heavier nuclei
- B. Heavy nucleus into lighter nuclei
- C. Electron into protons
- D. Photon into sound

Fast Route. Start from the defining relation or conservation statement and check units at the end.

Trap Watch. Keep frequency, wavelength, pressure, and speed in their own roles.

Answer. **B** Heavy nucleus into lighter nuclei

Worked Reasoning. Nuclear fission is the splitting of a heavy nucleus into lighter nuclei.

Why the other options are wrong.

- ◇ Option A: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Heavy nucleus into lighter nuclei.
- ◇ Option C: uses a qualitative statement that does not match the physical condition; it leads away from Heavy nucleus into lighter nuclei.
- ◇ Option D: confuses frequency, wavelength, speed, pressure, or conservation role; it leads away from Heavy nucleus into lighter nuclei.

Trick / Tip. The useful shortcut here is tied to nuclear composition: separate α , β , γ , protons, and neutrons by composition before choosing. The correct option should agree with Heavy nucleus into lighter nuclei, not just look numerically close.



Q158

Physics Mechanics

MCQ Focus. A 2 kg body moving at 10 m/s has kinetic energy:

- A. 20 J
- B. 50 J
- C. 100 J
- D. 200 J

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. C 100 J

Worked Reasoning. Kinetic energy $K = (\frac{1}{2})mv^2 = (\frac{1}{2})(2)(10^2) = 100$ J.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from 100 J.
- ◇ Option B: misses the square/proportionality relationship in the law; it leads away from 100 J.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from 100 J.

Trick / Tip. For this Mechanics pattern, notice energy relation: choose the conservation or energy formula named by the stem, not a nearby force relation. The correct option should agree with 100 J, not just look numerically close.

Q159

Physics Mechanics

MCQ Focus. If force is doubled and displacement is doubled in same direction, work becomes:

- A. Same
- B. Double
- C. Four times
- D. Half

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. ☒ C Four times

Worked Reasoning. Work $W=Fs$. Doubling both force and displacement makes work four times.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Four times.
- ◇ Option B: misses the square/proportionality relationship in the law; it leads away from Four times.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Four times.

Trick / Tip. This question rewards recognising force/work component: use $W=Fs \cos \theta$ and include only the component of force along displacement. The correct option should agree with Four times, not just look numerically close.



Q160

Physics Mechanics

MCQ Focus. An object is in equilibrium when net force is:

- A. Maximum
- B. Zero
- C. Equal to weight only
- D. Equal to friction only

Fast Route. Translate the sentence into the governing formula and use proportionality if exact calculation is not needed.

Trap Watch. Momentum, force, and energy are related but not interchangeable.

Answer. **B** Zero

Worked Reasoning. Equilibrium requires the net force to be zero.

Why the other options are wrong.

- ◇ Option A: uses a formula for a different mechanical quantity; it leads away from Zero.
- ◇ Option C: uses the right data but the wrong component or direction; it leads away from Zero.
- ◇ Option D: uses a formula for a different mechanical quantity; it leads away from Zero.

Trick / Tip. Question 160 is a net-force equilibrium item: equilibrium means the vector sum of forces is zero, not merely that two named forces are equal. The correct option should agree with Zero, not just look numerically close.

CHAPTER 3

English MCQ Strategy Lab

How to use this chapter. Use each page to practice exact meaning, grammar, and sentence logic. The correct answer must fit the prompt, not merely sound familiar.



Q161

English Vocabulary Meaning

MCQ Focus. Choose the closest meaning of 'aberration'.

- A. A normal habit
- B. A departure from what is normal
- C. A legal document
- D. A loud sound

Fast Route. Test the word in a simple sentence and choose the closest meaning, not the broadest association.

Trap Watch. Avoid words that are emotionally or contextually related but not true synonyms.

Answer. **B** A departure from what is normal

Worked Reasoning. Aberration means a departure from what is normal.

Why the other options are wrong.

- ◇ Option A: A normal habit does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from A departure from what is normal.
- ◇ Option C: A legal document does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from A departure from what is normal.
- ◇ Option D: A loud sound does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from A departure from what is normal.

Trick / Tip. The safe synonym test for aberration is substitution: the correct option should fit without changing the sentence sense.

Q162

English Vocabulary Meaning

MCQ Focus. Choose the closest meaning of 'candid'.

- A. Secretive
- B. Frank and honest
- C. Careless
- D. Temporary

Fast Route. Test the word in a simple sentence and choose the closest meaning, not the broadest association.

Trap Watch. Avoid words that are emotionally or contextually related but not true synonyms.

Answer. **B** Frank and honest

Worked Reasoning. Candid means frank and honest.

Why the other options are wrong.

- ◇ Option A: Secretive does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Frank and honest.
- ◇ Option C: Careless does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Frank and honest.
- ◇ Option D: Temporary does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Frank and honest.

Trick / Tip. For candid, make a quick personal sentence first; reject choices that share only the mood, not the meaning.



Q163

English Vocabulary Meaning

MCQ Focus. Choose the closest meaning of 'meticulous'.

- A. Careless
- B. Very careful and precise
- C. Angry
- D. Uncertain

Fast Route. Test the word in a simple sentence and choose the closest meaning, not the broadest association.

Trap Watch. Avoid words that are emotionally or contextually related but not true synonyms.

Answer. **B** Very careful and precise

Worked Reasoning. Meticulous means very careful and precise.

Why the other options are wrong.

- ◇ Option A: Careless does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Very careful and precise.
- ◇ Option C: Angry does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Very careful and precise.
- ◇ Option D: Uncertain does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Very careful and precise.

Trick / Tip. The safe synonym test for meticulous is substitution: the correct option should fit without changing the sentence sense.

Q164

English Vocabulary Meaning

MCQ Focus. Choose the closest meaning of 'tenacious'.

- A. Weak
- B. Persistent
- C. Noisy
- D. Brief

Fast Route. Test the word in a simple sentence and choose the closest meaning, not the broadest association.

Trap Watch. Avoid words that are emotionally or contextually related but not true synonyms.

Answer. **B** Persistent

Worked Reasoning. Tenacious means persistent and not easily giving up.

Why the other options are wrong.

- ◇ Option A: Weak does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Persistent.
- ◇ Option C: Noisy does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Persistent.
- ◇ Option D: Brief does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Persistent.

Trick / Tip. For tenacious, make a quick personal sentence first; reject choices that share only the mood, not the meaning.



Q165

English Vocabulary Meaning

MCQ Focus. Choose the closest meaning of 'pragmatic'.

- A. Practical
- B. Imaginary
- C. Ancient
- D. Musical

Fast Route. Test the word in a simple sentence and choose the closest meaning, not the broadest association.

Trap Watch. Avoid words that are emotionally or contextually related but not true synonyms.

Answer. A Practical

Worked Reasoning. Pragmatic means practical rather than theoretical.

Why the other options are wrong.

- ◇ Option B: Imaginary does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Practical.
- ◇ Option C: Ancient does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Practical.
- ◇ Option D: Musical does not carry the same core meaning demanded by exact synonym choice; it shifts the sense away from Practical.

Trick / Tip. The safe synonym test for pragmatic is substitution: the correct option should fit without changing the sentence sense.

Q166

English Vocabulary Antonym

MCQ Focus. Choose the antonym of 'scarce'.

- A. Rare
- B. Abundant
- C. Costly
- D. Hidden

Fast Route. Identify the core meaning first, then choose its direct opposite rather than a merely different word.

Trap Watch. A different idea is not always the opposite idea; look for direct reversal.

Answer. **B** Abundant

Worked Reasoning. The opposite of scarce is abundant.

Why the other options are wrong.

- ◇ Option A: Rare is not the direct reversal required here; the opposite meaning is expressed by Abundant.
- ◇ Option C: Costly is not the direct reversal required here; the opposite meaning is expressed by Abundant.
- ◇ Option D: Hidden is not the direct reversal required here; the opposite meaning is expressed by Abundant.

Trick / Tip. For scarce, write the core idea first, then reverse that idea exactly; do not settle for an unrelated word.



Q167

English Vocabulary Antonym

MCQ Focus. Choose the antonym of 'hostile'.

- A. Friendly
- B. Cruel
- C. Foreign
- D. Severe

Fast Route. Identify the core meaning first, then choose its direct opposite rather than a merely different word.

Trap Watch. A different idea is not always the opposite idea; look for direct reversal.

Answer. A Friendly

Worked Reasoning. The opposite of hostile is friendly.

Why the other options are wrong.

- ◇ Option B: Cruel is not the direct reversal required here; the opposite meaning is expressed by Friendly.
- ◇ Option C: Foreign is not the direct reversal required here; the opposite meaning is expressed by Friendly.
- ◇ Option D: Severe is not the direct reversal required here; the opposite meaning is expressed by Friendly.

Trick / Tip. For hostile, reward precision: choose the direct opposite of the core meaning, not a word from a different context.

Q168

English Vocabulary Antonym

MCQ Focus. Choose the antonym of 'opaque'.

- A. Transparent
- B. Dark
- C. Solid
- D. Heavy

Fast Route. Identify the core meaning first, then choose its direct opposite rather than a merely different word.

Trap Watch. A different idea is not always the opposite idea; look for direct reversal.

Answer. A Transparent

Worked Reasoning. The opposite of opaque is transparent.

Why the other options are wrong.

- ◇ Option B: Dark is not the direct reversal required here; the opposite meaning is expressed by Transparent.
- ◇ Option C: Solid is not the direct reversal required here; the opposite meaning is expressed by Transparent.
- ◇ Option D: Heavy is not the direct reversal required here; the opposite meaning is expressed by Transparent.

Trick / Tip. For opaque, write the core idea first, then reverse that idea exactly; do not settle for an unrelated word.

Q169

English Vocabulary Antonym

MCQ Focus. Choose the antonym of 'expand'.

- A. Extend
- B. Contract
- C. Explain
- D. Increase

Fast Route. Identify the core meaning first, then choose its direct opposite rather than a merely different word.

Trap Watch. A different idea is not always the opposite idea; look for direct reversal.

Answer. **B** Contract

Worked Reasoning. The opposite of expand is contract.

Why the other options are wrong.

- ◇ Option A: Extend is not the direct reversal required here; the opposite meaning is expressed by Contract.
- ◇ Option C: Explain is not the direct reversal required here; the opposite meaning is expressed by Contract.
- ◇ Option D: Increase is not the direct reversal required here; the opposite meaning is expressed by Contract.

Trick / Tip. For expand, reward precision: choose the direct opposite of the core meaning, not a word from a different context.

Q170

English Vocabulary Antonym

MCQ Focus. Choose the antonym of 'diligent'.

- A. Lazy
- B. Careful
- C. Active
- D. Honest

Fast Route. Identify the core meaning first, then choose its direct opposite rather than a merely different word.

Trap Watch. A different idea is not always the opposite idea; look for direct reversal.

Answer. A Lazy

Worked Reasoning. The opposite of diligent is lazy.

Why the other options are wrong.

- ◇ Option B: Careful is not the direct reversal required here; the opposite meaning is expressed by Lazy.
- ◇ Option C: Active is not the direct reversal required here; the opposite meaning is expressed by Lazy.
- ◇ Option D: Honest is not the direct reversal required here; the opposite meaning is expressed by Lazy.

Trick / Tip. For diligent, write the core idea first, then reverse that idea exactly; do not settle for an unrelated word.

Q171

English Analogy Logic

MCQ Focus. Analogy: Doctor : Hospital :: Teacher : ?

- A. Court
- B. School
- C. Laboratory
- D. Market

Fast Route. Name the relationship in the first pair, then demand the same relationship in the second pair.

Trap Watch. A topic match is weaker than a relationship match.

Answer. **B** School

Worked Reasoning. A doctor works in a hospital; a teacher works in a school.

Why the other options are wrong.

- ◇ Option A: Court breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.
- ◇ Option C: Laboratory breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.
- ◇ Option D: Market breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.

Trick / Tip. For the pair Doctor : Hospital, preserve the relationship before considering the topic of the options.

Q172

English Analogy Logic

MCQ Focus. Analogy: Pen : Writer :: Brush : ?

- A. Painter
- B. Farmer
- C. Driver
- D. Singer

Fast Route. Name the relationship in the first pair, then demand the same relationship in the second pair.

Trap Watch. A topic match is weaker than a relationship match.

Answer. A Painter

Worked Reasoning. A pen is used by a writer; a brush is used by a painter.

Why the other options are wrong.

- ◇ Option B: Farmer breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.
- ◇ Option C: Driver breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.
- ◇ Option D: Singer breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.

Trick / Tip. For the anchor pair Pen : Writer, say the relation as a sentence, then test the missing pair against it.

Q173

English Analogy Logic

MCQ Focus. Analogy: Thermometer : Temperature :: Barometer : ?

- A. Pressure
- B. Speed
- C. Current
- D. Length

Fast Route. Name the relationship in the first pair, then demand the same relationship in the second pair.

Trap Watch. A topic match is weaker than a relationship match.

Answer. A Pressure

Worked Reasoning. A thermometer measures temperature; a barometer measures pressure.

Why the other options are wrong.

- ◊ Option B: Speed breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.
- ◊ Option C: Current breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.
- ◊ Option D: Length breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.

Trick / Tip. For the pair Thermometer : Temperature, preserve the relationship before considering the topic of the options.

Q174

English Analogy Logic

MCQ Focus. Analogy: Seed : Plant :: Egg : ?

- A. Bird
- B. Nest
- C. Tree
- D. Soil

Fast Route. Name the relationship in the first pair, then demand the same relationship in the second pair.

Trap Watch. A topic match is weaker than a relationship match.

Answer. A Bird

Worked Reasoning. A seed develops into a plant; an egg develops into a bird.

Why the other options are wrong.

- ◇ Option B: Nest breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.
- ◇ Option C: Tree breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.
- ◇ Option D: Soil breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.

Trick / Tip. For the anchor pair Seed : Plant, say the relation as a sentence, then test the missing pair against it.

Q175

English Analogy Logic

MCQ Focus. Analogy: Book : Pages :: House : ?

- A. Rooms
- B. Pens
- C. Roads
- D. Clouds

Fast Route. Name the relationship in the first pair, then demand the same relationship in the second pair.

Trap Watch. A topic match is weaker than a relationship match.

Answer. A Rooms

Worked Reasoning. A book has pages; a house has rooms.

Why the other options are wrong.

- ◇ Option B: Pens breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.
- ◇ Option C: Roads breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.
- ◇ Option D: Clouds breaks the relationship pattern in the analogy; the second pair must match the first pair's relation.

Trick / Tip. For the pair Book : Pages, preserve the relationship before considering the topic of the options.

Q176

English

Spelling Accuracy

MCQ Focus. Choose the correctly spelled word.

- A. Accomodate
- B. Accommodate
- C. Accommodate
- D. Acomodate

Fast Route. Check the common double-letter or vowel trap; pronunciation is not enough.

Trap Watch. The most familiar misspelling is often placed beside the correct spelling.

Answer. ☒ Accommodate

Worked Reasoning. The correct spelling is accommodate, with double c and double m.

Why the other options are wrong.

- ◇ Option A: Accomodate is a distractor spelling; compare its doubled letters and vowel order with the correct form Accommodate.
- ◇ Option B: Accommodate is a distractor spelling; compare its doubled letters and vowel order with the correct form Accommodate.
- ◇ Option D: Acomodate is a distractor spelling; compare its doubled letters and vowel order with the correct form Accommodate.

Trick / Tip. For spelling Accommodate, mark doubled consonants or vowel order before reading the options; the distractors imitate common misspellings.



Q177

English Spelling Accuracy

MCQ Focus. Choose the correctly spelled word.

- A. Occasion
- B. Ocassion
- C. Occassion
- D. Ocaasion

Fast Route. Check the common double-letter or vowel trap; pronunciation is not enough.

Trap Watch. The most familiar misspelling is often placed beside the correct spelling.

Answer. A Occasion

Worked Reasoning. The correct spelling is occasion, with double c and one s.

Why the other options are wrong.

- ◇ Option B: Ocassion is a distractor spelling; compare its doubled letters and vowel order with the correct form Occasion.
- ◇ Option C: Occassion is a distractor spelling; compare its doubled letters and vowel order with the correct form Occasion.
- ◇ Option D: Ocaasion is a distractor spelling; compare its doubled letters and vowel order with the correct form Occasion.

Trick / Tip. Spelling Occasion is visual work: compare letter groups slowly rather than relying on how the word sounds.

Q178

English

Spelling Accuracy

MCQ Focus. Choose the correctly spelled word.

- A. Seperate
- B. Separate
- C. Seperete
- D. Separate

Fast Route. Check the common double-letter or vowel trap; pronunciation is not enough.

Trap Watch. The most familiar misspelling is often placed beside the correct spelling.

Answer. **B** Separate

Worked Reasoning. The correct spelling is separate.

Why the other options are wrong.

- ◇ Option A: Seperate is a distractor spelling; compare its doubled letters and vowel order with the correct form Separate.
- ◇ Option C: Seperete is a distractor spelling; compare its doubled letters and vowel order with the correct form Separate.
- ◇ Option D: Separate is a distractor spelling; compare its doubled letters and vowel order with the correct form Separate.

Trick / Tip. For spelling Separate, mark doubled consonants or vowel order before reading the options; the distractors imitate common misspellings.



Q179

English Spelling Accuracy

MCQ Focus. Choose the correctly spelled word.

- A. Conscience
- B. Consciense
- C. Consience
- D. Consciencee

Fast Route. Check the common double-letter or vowel trap; pronunciation is not enough.

Trap Watch. The most familiar misspelling is often placed beside the correct spelling.

Answer. A Conscience

Worked Reasoning. Conscience is the correct spelling for the moral sense of right and wrong.

Why the other options are wrong.

- ◇ Option B: Consciense is a distractor spelling; compare its doubled letters and vowel order with the correct form Conscience.
- ◇ Option C: Consience is a distractor spelling; compare its doubled letters and vowel order with the correct form Conscience.
- ◇ Option D: Consciencee is a distractor spelling; compare its doubled letters and vowel order with the correct form Conscience.

Trick / Tip. Spelling Conscience is visual work: compare letter groups slowly rather than relying on how the word sounds.

Q180

English

Spelling Accuracy

MCQ Focus. Choose the correctly spelled word.

- A. Definately
- B. Definitely
- C. Definitly
- D. Definetely

Fast Route. Check the common double-letter or vowel trap; pronunciation is not enough.

Trap Watch. The most familiar misspelling is often placed beside the correct spelling.

Answer. **B** Definitely

Worked Reasoning. Definitely is the correct spelling.

Why the other options are wrong.

- ◇ Option A: Definately is a distractor spelling; compare its doubled letters and vowel order with the correct form Definitely.
- ◇ Option C: Definitly is a distractor spelling; compare its doubled letters and vowel order with the correct form Definitely.
- ◇ Option D: Definetely is a distractor spelling; compare its doubled letters and vowel order with the correct form Definitely.

Trick / Tip. For spelling Definitely, mark doubled consonants or vowel order before reading the options; the distractors imitate common misspellings.



Q181

English

Grammar Fit

MCQ Focus. Fill in the blank: He is senior me.

- A. than
- B. from
- C. to
- D. with

Fast Route. Insert each option into the sentence and test idiom, article, or preposition fit.

Trap Watch. Prepositions and articles are idiomatic; a literal translation often creates the trap.

Answer. C to

Worked Reasoning. The adjective senior takes the preposition to.

Why the other options are wrong.

- ◇ Option A: than does not fit the sentence idiom or grammar pattern; the blank requires to.
- ◇ Option B: from does not fit the sentence idiom or grammar pattern; the blank requires to.
- ◇ Option D: with does not fit the sentence idiom or grammar pattern; the blank requires to.

Trick / Tip. In the blank sentence He is senior me., test idiom first; the shortest option is not safer unless it fits the exact phrase.



Q182

English Grammar Fit

MCQ Focus. Fill in the blank: She insisted paying the bill.

- A. on
- B. at
- C. for
- D. to

Fast Route. Insert each option into the sentence and test idiom, article, or preposition fit.

Trap Watch. Prepositions and articles are idiomatic; a literal translation often creates the trap.

Answer. **A** on

Worked Reasoning. The verb insisted is followed by on.

Why the other options are wrong.

- ◇ Option B: at does not fit the sentence idiom or grammar pattern; the blank requires on.
- ◇ Option C: for does not fit the sentence idiom or grammar pattern; the blank requires on.
- ◇ Option D: to does not fit the sentence idiom or grammar pattern; the blank requires on.

Trick / Tip. For the blank sentence She insisted paying the bill., read the whole sentence with each option inserted; the correct option must satisfy grammar.



Q183

English Grammar Fit

MCQ Focus. Fill in the blank: The committee has reached decision.

- A. a
- B. an
- C. the
- D. no article

Fast Route. Insert each option into the sentence and test idiom, article, or preposition fit.

Trap Watch. Prepositions and articles are idiomatic; a literal translation often creates the trap.

Answer. **A** a

Worked Reasoning. The phrase is reached a decision, using the article a.

Why the other options are wrong.

- ◇ Option B: an does not fit the sentence idiom or grammar pattern; the blank requires a.
- ◇ Option C: the does not fit the sentence idiom or grammar pattern; the blank requires a.
- ◇ Option D: no article does not fit the sentence idiom or grammar pattern; the blank requires a.

Trick / Tip. In the blank sentence The committee has reached decision., test idiom first; the shortest option is not safer unless it fits the exact phrase.

Q184

English Sentence Correction

MCQ Focus. Choose the correct sentence.

- A. Neither of the answers are correct.
- B. Neither of the answers is correct.
- C. Neither of answers is correct.
- D. Neither answers are correct.

Fast Route. Check subject-verb agreement, tense, article use, and phrase structure in that order.

Trap Watch. Plural-looking subjects and collective nouns often mislead agreement checks.

Answer. **B** Neither of the answers is correct.

Worked Reasoning. Neither is singular here, so the verb should be is.

Why the other options are wrong.

- ◇ Option A: this sentence keeps a grammar fault in agreement, tense, or structure, while the correct sentence is option B.
- ◇ Option C: this sentence keeps a grammar fault in agreement, tense, or structure, while the correct sentence is option B.
- ◇ Option D: this sentence keeps a grammar fault in agreement, tense, or structure, while the correct sentence is option B.

Trick / Tip. For this sentence, test subject-verb agreement first, then tense and article use; this order catches most traps.



Q185

English Sentence Correction

MCQ Focus. Choose the correct sentence.

- A. The data is analyzed yesterday.
- B. The data were analyzed yesterday.
- C. The data analyzing yesterday.
- D. The data was analyze yesterday.

Fast Route. Check subject-verb agreement, tense, article use, and phrase structure in that order.

Trap Watch. Plural-looking subjects and collective nouns often mislead agreement checks.

Answer. **B** The data were analyzed yesterday.

Worked Reasoning. In this formal sentence, data is treated as plural and the past passive form is were analyzed.

Why the other options are wrong.

- ◇ Option A: this sentence keeps a grammar fault in agreement, tense, or structure, while the correct sentence is option B.
- ◇ Option C: this sentence keeps a grammar fault in agreement, tense, or structure, while the correct sentence is option B.
- ◇ Option D: this sentence keeps a grammar fault in agreement, tense, or structure, while the correct sentence is option B.

Trick / Tip. For this sentence-correction item, choose the option that is grammatically complete, not merely conversational.



Q186

English Reported Speech

MCQ Focus. Choose the correct indirect speech: He said, 'I am tired.'

- A. He said that I am tired.
- B. He said that he was tired.
- C. He says that he was tired.
- D. He said he is tiring.

Fast Route. Shift pronoun and tense according to the reporting verb and speaker reference.

Trap Watch. Do not keep the original speaker pronoun if the reporting sentence requires a shift.

Answer. **B** He said that he was tired.

Worked Reasoning. In indirect speech, I changes to he and am changes to was.

Why the other options are wrong.

- ◇ Option A: it mishandles pronoun or tense shift in reported speech; option B preserves the reporting structure.
- ◇ Option C: it mishandles pronoun or tense shift in reported speech; option B preserves the reporting structure.
- ◇ Option D: it mishandles pronoun or tense shift in reported speech; option B preserves the reporting structure.

Trick / Tip. For reported speech, change pronouns from the speaker viewpoint to the reporter viewpoint, then backshift tense if needed.

Q187

English Voice Conversion

MCQ Focus. Choose the correct passive voice: They completed the project.

- A. The project completed them.
- B. The project was completed by them.
- C. The project is completed by them.
- D. The project has completed.

Fast Route. Move the object into subject position and preserve tense through the correct be-form.

Trap Watch. Do not change tense while changing voice.

Answer. **B** The project was completed by them.

Worked Reasoning. Passive voice uses object + was/were + past participle: the project was completed by them.

Why the other options are wrong.

- ◇ Option A: it does not preserve the active sentence in proper passive form; option B keeps tense and object movement correct.
- ◇ Option C: it does not preserve the active sentence in proper passive form; option B keeps tense and object movement correct.
- ◇ Option D: it does not preserve the active sentence in proper passive form; option B keeps tense and object movement correct.

Trick / Tip. Passive voice here is structure work: object plus be-form plus past participle, with tense preserved.

Q188

English Error Identification

MCQ Focus. Identify the error: Each of the students have submitted the form.

- A. Each
- B. students
- C. have
- D. form

Fast Route. Find the one word or phrase that breaks agreement, idiom, or grammatical structure.

Trap Watch. The error may be a short phrase, not the longest option.

Answer. C have

Worked Reasoning. Each is singular, so have should be replaced by has.

Why the other options are wrong.

- ◇ Option A: this part is not the fault being tested; the actual error is the phrase identified by option C.
- ◇ Option B: this part is not the fault being tested; the actual error is the phrase identified by option C.
- ◇ Option D: this part is not the fault being tested; the actual error is the phrase identified by option C.

Trick / Tip. For this error item, inspect small grammar words carefully; preposition and agreement errors are often shorter than vocabulary errors.



Q189

English Error Identification

MCQ Focus. Identify the error: He is good in mathematics.

- A. He
- B. is
- C. good in
- D. mathematics

Fast Route. Find the one word or phrase that breaks agreement, idiom, or grammatical structure.

Trap Watch. The error may be a short phrase, not the longest option.

Answer. C good in

Worked Reasoning. The standard phrase is good at mathematics, not good in mathematics.

Why the other options are wrong.

- ◊ Option A: this part is not the fault being tested; the actual error is the phrase identified by option C.
- ◊ Option B: this part is not the fault being tested; the actual error is the phrase identified by option C.
- ◊ Option D: this part is not the fault being tested; the actual error is the phrase identified by option C.

Trick / Tip. In this error item, do not rewrite the sentence; locate the exact faulty phrase.

Q190

English Logical Connectors

MCQ Focus. Choose the best connector: He studied hard; , he failed the test.

- A. therefore
- B. however
- C. because
- D. so

Fast Route. Choose the connector that matches contrast, cause, result, or continuation.

Trap Watch. Therefore, however, because, and so mark different logic; choose from meaning, not habit.

Answer. **B** however

Worked Reasoning. However shows contrast between studying hard and failing.

Why the other options are wrong.

- ◇ Option A: therefore gives the wrong logical link between the two clauses; the sentence needs however.
- ◇ Option C: because gives the wrong logical link between the two clauses; the sentence needs however.
- ◇ Option D: so gives the wrong logical link between the two clauses; the sentence needs however.

Trick / Tip. For this connector item, decide whether the second clause contrasts, explains, or results from the first before seeing the options.



Q191

English Context Completion

MCQ Focus. Choose the best sentence completion: The experiment was repeated to the results.

- A. verify
- B. neglect
- C. confuse
- D. erase

Fast Route. Use sentence meaning to choose the word that completes the thought without contradiction.

Trap Watch. A positive-sounding word can be wrong if it contradicts the sentence purpose.

Answer. A verify

Worked Reasoning. To verify results means to check or confirm them.

Why the other options are wrong.

- ◇ Option B: neglect does not complete the sentence meaning cleanly; it changes the intended context away from verify.
- ◇ Option C: confuse does not complete the sentence meaning cleanly; it changes the intended context away from verify.
- ◇ Option D: erase does not complete the sentence meaning cleanly; it changes the intended context away from verify.

Trick / Tip. This completion item is a context test. Remove any option that makes the sentence internally inconsistent.

Q192

English Context Completion

MCQ Focus. Choose the best sentence completion: His explanation was so that no one asked another question.

- A. ambiguous
- B. clear
- C. irrelevant
- D. fragile

Fast Route. Use sentence meaning to choose the word that completes the thought without contradiction.

Trap Watch. A positive-sounding word can be wrong if it contradicts the sentence purpose.

Answer. **B** clear

Worked Reasoning. If no one asked another question, the explanation was clear.

Why the other options are wrong.

- ◇ Option A: ambiguous does not complete the sentence meaning cleanly; it changes the intended context away from clear.
- ◇ Option C: irrelevant does not complete the sentence meaning cleanly; it changes the intended context away from clear.
- ◇ Option D: fragile does not complete the sentence meaning cleanly; it changes the intended context away from clear.

Trick / Tip. For this completion, identify the sentence goal first; the correct word must support that goal without reversing tone.



Q193

English Context Completion

MCQ Focus. Choose the best sentence completion: The manager tried to the conflict before it became serious.

- A. escalate
- B. resolve
- C. ignore
- D. invent

Fast Route. Use sentence meaning to choose the word that completes the thought without contradiction.

Trap Watch. A positive-sounding word can be wrong if it contradicts the sentence purpose.

Answer. **B** resolve

Worked Reasoning. To resolve a conflict means to settle it.

Why the other options are wrong.

- ◇ Option A: escalate does not complete the sentence meaning cleanly; it changes the intended context away from resolve.
- ◇ Option C: ignore does not complete the sentence meaning cleanly; it changes the intended context away from resolve.
- ◇ Option D: invent does not complete the sentence meaning cleanly; it changes the intended context away from resolve.

Trick / Tip. This completion item is a context test. Remove any option that makes the sentence internally inconsistent.

Q194

English Context Completion

MCQ Focus. Choose the best sentence completion: A scientist must be when recording observations.

- A. careless
- B. accurate
- C. hostile
- D. vague

Fast Route. Use sentence meaning to choose the word that completes the thought without contradiction.

Trap Watch. A positive-sounding word can be wrong if it contradicts the sentence purpose.

Answer. **B** accurate

Worked Reasoning. A scientist must be accurate when recording observations.

Why the other options are wrong.

- ◇ Option A: careless does not complete the sentence meaning cleanly; it changes the intended context away from accurate.
- ◇ Option C: hostile does not complete the sentence meaning cleanly; it changes the intended context away from accurate.
- ◇ Option D: vague does not complete the sentence meaning cleanly; it changes the intended context away from accurate.

Trick / Tip. For this completion, identify the sentence goal first; the correct word must support that goal without reversing tone.

Q195

English Reading Inference

MCQ Focus. Reading: A short passage says that excessive screen use before sleep can reduce sleep quality because bright light delays melatonin release. What is the main idea?

- A. Screens always improve sleep
- B. Bright screens may disturb sleep
- C. Melatonin is unrelated to sleep
- D. Sleep depends only on diet

Fast Route. Use only the passage evidence; do not add outside assumptions.

Trap Watch. Passage questions punish outside knowledge; stay inside the given statement.

Answer. **B** Bright screens may disturb sleep

Worked Reasoning. The passage says bright screen use before sleep can reduce sleep quality.

Why the other options are wrong.

- ◇ Option A: Screens always improve sleep is not supported by the passage statement; the evidence points to Bright screens may disturb sleep.
- ◇ Option C: Melatonin is unrelated to sleep is not supported by the passage statement; the evidence points to Bright screens may disturb sleep.
- ◇ Option D: Sleep depends only on diet is not supported by the passage statement; the evidence points to Bright screens may disturb sleep.

Trick / Tip. This reading MCQ should be answered from evidence, not memory; choose only what the passage states or strongly implies.



Q196

English Reading Inference

MCQ Focus. Reading: A passage says the Indus Valley cities had planned streets and drainage. Which conclusion is most supported?

- A. They lacked organization
- B. They had urban planning
- C. They used no water
- D. They were all modern cities

Fast Route. Use only the passage evidence; do not add outside assumptions.

Trap Watch. Passage questions punish outside knowledge; stay inside the given statement.

Answer. **B** They had urban planning

Worked Reasoning. Planned streets and drainage support the conclusion that the cities had urban planning.

Why the other options are wrong.

- ◇ Option A: They lacked organization is not supported by the passage statement; the evidence points to They had urban planning.
- ◇ Option C: They used no water is not supported by the passage statement; the evidence points to They had urban planning.
- ◇ Option D: They were all modern cities is not supported by the passage statement; the evidence points to They had urban planning.

Trick / Tip. For this reading item, underline the exact phrase in the passage that supports the answer and ignore outside information.



Q197

English Reading Inference

MCQ Focus. Reading: A passage says renewable energy reduces dependence on fossil fuels but needs storage solutions. The tone is best described as:

- A. Balanced
- B. Mocking
- C. Angry
- D. Completely hopeless

Fast Route. Use only the passage evidence; do not add outside assumptions.

Trap Watch. Passage questions punish outside knowledge; stay inside the given statement.

Answer. A Balanced

Worked Reasoning. The passage gives both a benefit and a limitation, so the tone is balanced.

Why the other options are wrong.

- ◇ Option B: Mocking is not supported by the passage statement; the evidence points to Balanced.
- ◇ Option C: Angry is not supported by the passage statement; the evidence points to Balanced.
- ◇ Option D: Completely hopeless is not supported by the passage statement; the evidence points to Balanced.

Trick / Tip. This reading MCQ should be answered from evidence, not memory; choose only what the passage states or strongly implies.

Q198

English Reading Inference

MCQ Focus. Reading: A passage says exercise improves circulation, mood, and stamina. Which option is not mentioned?

- A. Better circulation
- B. Improved mood
- C. Higher stamina
- D. Permanent immunity

Fast Route. Use only the passage evidence; do not add outside assumptions.

Trap Watch. Passage questions punish outside knowledge; stay inside the given statement.

Answer. **D** Permanent immunity

Worked Reasoning. The passage mentions circulation, mood, and stamina, but not permanent immunity.

Why the other options are wrong.

- ◇ Option A: Better circulation is not supported by the passage statement; the evidence points to Permanent immunity.
- ◇ Option B: Improved mood is not supported by the passage statement; the evidence points to Permanent immunity.
- ◇ Option C: Higher stamina is not supported by the passage statement; the evidence points to Permanent immunity.

Trick / Tip. For this reading item, underline the exact phrase in the passage that supports the answer and ignore outside information.



Q199

English Phrase Replacement

MCQ Focus. Choose the word that best replaces the underlined idea: 'He spoke without preparation.'

- A. impromptu
- B. reluctantly
- C. silently
- D. formally

Fast Route. Replace the underlined idea with the single option that preserves the original meaning.

Trap Watch. Do not choose an adverb that changes manner, tone, or timing of the original phrase.

Answer. A impromptu

Worked Reasoning. Impromptu means done or spoken without preparation.

Why the other options are wrong.

- ◇ Option B: reluctantly changes the manner of the action instead of preserving the idea in the underlined phrase; the phrase is best replaced by impromptu.
- ◇ Option C: silently changes the manner of the action instead of preserving the idea in the underlined phrase; the phrase is best replaced by impromptu.
- ◇ Option D: formally changes the manner of the action instead of preserving the idea in the underlined phrase; the phrase is best replaced by impromptu.

Trick / Tip. This replacement item is not asking for mood or formality; choose the word that keeps the original idea intact.



Q200

English One-Word Substitution

MCQ Focus. Choose the most appropriate one-word substitution: A person who studies earthquakes is a:

- A. Geologist
- B. Seismologist
- C. Meteorologist
- D. Biologist

Fast Route. Match the specialist term to the definition, not to a related field.

Trap Watch. Related sciences or professions are not interchangeable.

Answer. **B** Seismologist

Worked Reasoning. A seismologist studies earthquakes.

Why the other options are wrong.

- ◇ Option A: Geologist names a related but different role; the definition points specifically to Seismologist.
- ◇ Option C: Meteorologist names a related but different role; the definition points specifically to Seismologist.
- ◇ Option D: Biologist names a related but different role; the definition points specifically to Seismologist.

Trick / Tip. For this one-word substitution, connect the definition to the specialist role; related fields are distractors if the exact profession differs.



Appendices



APPENDIX A

Final Test - Reserved

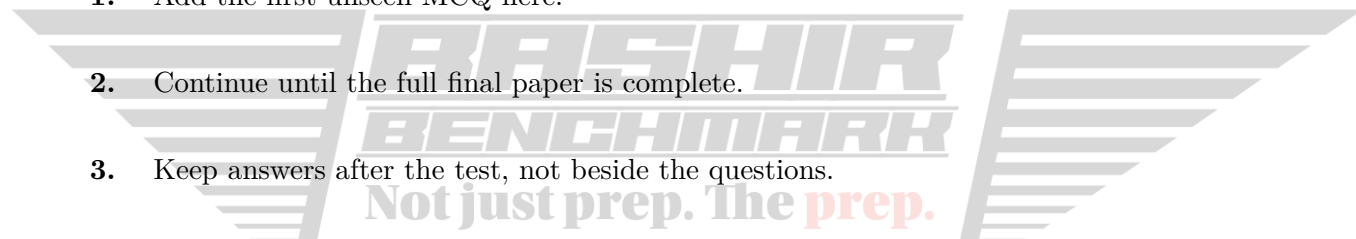
This section is intentionally reserved for the final unseen paper. Add the real final test here after the learner completes the MCQ strategy chapters.

Final Test Placeholder

1. Add the first unseen MCQ here.

2. Continue until the full final paper is complete.

3. Keep answers after the test, not beside the questions.



APPENDIX B

Quick Answer Key for the Strategy Chapters

1. A	14. B	27. B	40. A
2. B	15. A	28. B	41. B
3. A	16. A	29. A	42. C
4. A	17. A	30. B	43. D
5. A	18. C	31. B	44. C
6. B	19. B	32. B	45. C
7. B	20. C	33. C	46. B
8. A	21. C	34. B	47. B
9. A	22. A	35. B	48. C
10. B	23. A	36. B	49. C
11. A	24. C	37. C	50. C
12. A	25. B	38. B	51. B
13. C	26. A	39. B	52. B

Appendix B. Quick Answer Key for
the Strategy Chapters

53. C	73. C	93. B	113. C
54. A	74. B	94. B	114. B
55. B	75. C	95. B	115. A
56. A	76. A	96. C	116. B
57. A	77. A	97. C	117. B
58. B	78. C	98. A	118. B
59. C	79. A	99. C	119. C
60. C	80. C	100. A	120. B
61. B	81. B	101. C	121. B
62. B	82. A	102. A	122. C
63. C	83. B	103. B	123. C
64. C	84. D	104. B	124. B
65. B	85. B	105. B	125. C
66. B	86. A	106. C	126. B
67. B	87. C	107. C	127. A
68. A	88. A	108. B	128. C
69. A	89. C	109. C	129. A
70. C	90. C	110. A	130. C
71. A	91. C	111. B	131. B
72. B	92. B	112. C	132. B



Appendix B. Quick Answer Key for
the Strategy Chapters

133. D	150. C	167. A	184. B
134. B	151. A	168. A	185. B
135. B	152. B	169. B	186. B
136. B	153. B	170. A	187. B
137. B	154. C	171. B	188. C
138. B	155. B	172. A	189. C
139. A	156. B	173. A	190. B
140. A	157. B	174. A	191. A
141. B	158. C	175. A	192. B
142. B	159. C	176. C	193. B
143. B	160. B	177. A	194. B
144. A	161. B	178. B	195. B
145. B	162. B	179. A	196. B
146. B	163. B	180. B	197. A
147. A	164. B	181. C	198. D
148. B	165. A	182. A	199. A
149. B	166. B	183. A	200. B



APPENDIX C

Personal Trick Register

Use this register for traps you personally missed.

Q No.	Topic	Missed Trick	New Rule to Remember
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			

Appendix B. Quick Answer Key for
the Strategy Chapters

Q No.	Topic	Missed Trick	New Rule to Remember
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			
26			
27			



Appendix B. Quick Answer Key for
the Strategy Chapters

Q No.	Topic	Missed Trick	New Rule to Remember
28			
29			
30			
31			
32			
33			
34			
35			
36			
37			
38			
39			
40			
41			
42			
43			



Appendix B. Quick Answer Key for
the Strategy Chapters

Q No.	Topic	Missed Trick	New Rule to Remember
44			
45			
46			
47			
48			
49			
50			
51			
52			
53			
54			
55			
56			
57			
58			
59			



Appendix B. Quick Answer Key for
the Strategy Chapters

Q No.	Topic	Missed Trick	New Rule to Remember
60			

